CALLOW SCHOOL · THE TEXTBOOK

INTRODUCTORY SERIES



PRINCIPLES of Mind

An Introduction in Seven Floors

THE FACULTY OF CALLOW SCHOOL

First Edition

CALLOW SCHOOL PRESS

DAEMONS · for release v11.337.4
every word quoted from the game is a DRAFT

PRINCIPLES OF MIND

An Introduction in Seven Floors

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An Introduction in Seven Floors

FIRST EDITION

The Faculty of Callow School

with the boards of all seven floors

CALLOW SCHOOL PRESS

CALLOW CITY

Principles of Mind: An Introduction in Seven Floors. First edition. This is CALLOW SCHOOL's TEXTBOOK, from the game DAEMONS, written out in full as a university textbook would be. Callow School, its Press and its Faculty belong to the game.

The research in this book is real. Every study, date and finding it reports comes from the published literature listed in the References at the back, and was checked against it.

Every passage quoted from the game, the floors' placards and boards, is reproduced word for word from the build, and like every word in the game it is a DRAFT.

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For every student who was told it was hard,
when it was only unfamiliar.

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Preface

To the Student

This book is an introduction to the scientific study of the mind. It is written for a reader who has never taken a course in the subject and who wants to understand, carefully and from the evidence, what happens to a piece of information between the moment it reaches a person and the moment it changes what that person does next.

The book follows the building it was written for. Callow School teaches the mind on seven floors, one subject to a floor, and each floor's teacher writes the lesson on a board in five short parts. Those thirty-five board entries are deliberately brief: a board is something to read from the back of a room. This textbook keeps every one of them, word for word, and then does what a board cannot. It explains each idea, shows where it came from, reports the research that established it, and marks the places where the evidence is more complicated than a sentence in chalk can be.

The title is a small act of homage. William James called his great survey of 1890 *The Principles of Psychology*, and it opens its chapter on attention with the remark that "everyone knows what attention is." This book shares his conviction that the things everyone already knows about their own minds are exactly the things most worth examining.

What This Book Teaches

The seven chapters follow the order of the floors, and the order is not arbitrary. Read upward, the building traces what happens to a piece of information: it is **named**, **noticed**, **kept**, **sorted**, **changed**, **bent** and **corrected**.

Chapter 1, Language, comes first because everything else is taught in words. It examines how a word points at a thing, how words are built from smaller meaningful pieces, how order carries meaning, how context settles which of several meanings a word carries, and why English spelling is so irregular.

Chapter 2, Attention, concerns what gets processed at all. Attention is limited, it can be directed in advance, it finds some things effortlessly and others only by search, it switches at a cost, and it can miss what is in plain view.

Chapter 3, Memory, follows what is kept. It treats memory as reconstruction rather than recording, and covers forgetting, interference, the reshaping of a memory each time it is recalled, and the robust advantage of spacing study over time.

Chapter 4, Categories, concerns how the mind sorts. Real categories have central members and blurred edges, are organised at several levels, depend on which features are attended to, and always leave a remainder.

Chapter 5, Learning, describes how experience changes the brain itself: cells joined at synapses whose strength changes with use, thresholds that decide whether a cell fires, the indispensable role of feedback, and the pruning of connections during development.

Chapter 6, Bias, catalogues the systematic ways judgment departs from what the evidence warrants: framing, anchoring, confirmation, hindsight, and the informal fallacies of argument.

Chapter 7, Error, closes the sequence with the idea that learning is driven by the gap between what was expected and what occurred. It treats surprise, correction, the size of each step, the value of noticed mistakes, and the cost of certainty.

A Conclusion draws the chapters together and turns their findings into advice for studying, including for the examination the school sets. A Glossary and a list of References close the book.

How to Use This Book

Each chapter has the same parts, so that a reader always knows where to look.

- **The placard.** Every chapter opens with the placard that stands by its floor's stairs, stating in two sentences what the floor is for.
- **Learning objectives.** A short list of what you should be able to do when you have finished the chapter. Return to it afterwards and test yourself against it.
- **On the board.** Each section begins with one of the five board entries, quoted exactly and set on slate. *This is what the school's examination covers.* The text that follows the

board explains it; the board itself is what to study.

- **Key terms** are printed in **bold** where they are defined, and collected in the Glossary with the section in which each appears.
- **Research Spotlight** boxes describe a single landmark study in detail: who did it, what they did, what they found, and why it mattered.
- **Try It** boxes give a short exercise you can do with a pencil or a friend, so that you experience a finding rather than only read about it.
- **A Closer Look** boxes mark places where the evidence is contested, or where a popular version of a finding goes further than the research does.
- **Chapter review.** Each chapter ends with a summary, its key terms, review questions and suggestions for further reading.

A NOTE ON THE BOARDS

The boards and placards are quoted from the game exactly as they are written in the build, and like every word in the game they are DRAFT. Where a board simplifies, as boards must, this book says so and gives the fuller account. Where the two seem to disagree, study the board for the examination and the text for understanding; Chapter 7 will explain why the second is worth more.

CHAPTER I

Language

THE PLACARD BY THE STAIRS · IF

What a word points at, how words come apart, and why their order matters.

Everything else here is taught in words, so words come first.

The syllabus is on the desk.

Learning Objectives

1. Distinguish a word from the thing it refers to, and explain why the link between them is arbitrary.
2. Identify the morphemes in a complex word and use them to interpret words you have never seen.
3. Explain how word order carries meaning in English, and how other languages carry the same information.
4. Describe how listeners resolve lexical and structural ambiguity, and how quickly they do so.
5. Explain why English spelling is irregular, and what that irregularity costs a learner.

Language is the first subject in the building for a practical reason, which the placard states plainly: everything else is taught in words. Before a student can be told about attention or memory, the student needs words for them, and needs to understand what words are and how they work.

The scientific study of language and the mind, **psycholinguistics**, treats a language as a layered system. Sounds combine into meaningful pieces; pieces combine into words; words combine, according to rules, into sentences; and sentences mean things in context. The linguist Charles Hockett (1960) listed the "design features" that set human language

apart from other systems of communication. Three of them organise this chapter: **arbitrariness** (the form of a word does not resemble its meaning), **duality of patterning** (meaningless units such as sounds and letters combine into meaningful ones), and **productivity** (a finite set of pieces and rules yields an unlimited number of new expressions).

1.1 Names

ON THE BOARD · IF LANGUAGE · **NAMES**

A name is a handle.

It lets you point at a thing without having to carry it.

Rename a thing and it stays exactly what it was.

Only the handle moved.

A word is a **sign**, and the Swiss linguist Ferdinand de Saussure, in lectures published after his death as the *Course in General Linguistics* (1916), divided the sign into two parts: the **signifier**, the sound or written form, and the **signified**, the concept it calls up. His central claim was that the bond between the two is **arbitrary**. Nothing about the sounds of *tree* resembles a tree, which is why the same concept is *arbre* in French and *Baum* in German. The few exceptions, such as **onomatopoeia** (*buzz*, *hiss*), are fewer and less consistent across languages than one might expect.

Because the bond is arbitrary, a name works exactly as the board says: as a handle. It allows us to refer to a thing in its absence, to talk about it, plan around it and pass it to someone else, without the thing being present. And because the bond is conventional, the handle can be changed without changing the thing. Shakespeare's Juliet makes the point in a single line: "a rose by any other name would smell as sweet."

Philosophers sharpened the idea. Gottlob Frege (1892) distinguished a name's **reference**, the object it picks out, from its **sense**, the way it presents that object. The "morning star" and the "evening star" have different senses but the same reference: both are the planet Venus. Discovering that they named one object was a discovery about the sky, not about the words. Renaming changes the sense, the route by which we reach a thing; it does not change the thing reached.

RESEARCH SPOTLIGHT · A NAME CHANGES THE MEMORY, NOT THE THING

Carmichael, Hogan and Walter (1932) showed participants simple, ambiguous line drawings, each introduced with one of two names. The same drawing was called "eyeglasses" for some participants and "dumbbells" for others. Asked later to reproduce the drawings from memory, participants tended to draw them to resemble the name they had been given. The drawings themselves had not changed at all; what changed was what people kept of them. The finding refines the board: renaming a thing leaves it exactly what it was, but a name can still shape what a mind stores about it, a theme Chapter 3 takes up in detail.

1.2 Pieces

ON THE BOARD · IF LANGUAGE · **PIECES**

Words come apart.

UN-READ-ABLE is three pieces. Each one means something alone.

Learn the pieces, and you can read words nobody taught you.

The smallest unit of language that carries meaning is the **morpheme**. *Unreadable* contains three: the prefix *un-* ("not"), the root *read*, and the suffix *-able* ("can be done"). Morphemes that can stand alone as words, like *read*, are **free morphemes**; those that cannot, like *un-* and *-able*, are **bound morphemes**.

Linguists distinguish two kinds of bound morpheme. **Inflectional morphemes** mark grammatical information without changing a word's basic category: the *-s* of *daemons*, the *-ed* of *walked*. **Derivational morphemes** build new words, often of a new category: *-able* turns the verb *read* into the adjective *readable*, and *-ness* turns that into the noun *readableness*.

The board's promise, that knowing the pieces lets you read words nobody taught you, is the property Hockett called productivity, and it is one of the best-established facts in the study of children's language.

RESEARCH SPOTLIGHT · THE WUG TEST

Jean Berko (1958) showed young children a drawing of a small bird-like creature and said, "This is a wug." She then showed two of them: "Now there are two of them. There are two..." Children readily answered "wugs", pronouncing the plural ending correctly, although they could never have heard the word before, because Berko had invented it. They did the same with invented verbs ("he ricks every day; yesterday he...") and invented adjectives. The children were not repeating forms they had memorised. They had learned the rules for combining pieces, and applied them to a word that was new.

Morphology matters well beyond the nursery. The written English that schoolchildren meet contains far more distinct words than could ever be taught one at a time; Nagy and Anderson (1984) estimated the number of distinct word families in printed school English at roughly 88,500. Most of the vocabulary a reader acquires is therefore learned indirectly, and much of it by recognising known pieces in unfamiliar combinations. Readers who are aware of morphological structure tend to learn new words more readily.

TRY IT

Divide each word into its morphemes and say what each piece contributes: *misremembered*, *rethinkable*, *unforgettably*, *predictable*. Then invent a word that no dictionary contains, built only from real pieces (for example *unrecategorisable*), and ask a friend what it means. If they can tell you, you have just observed productivity at work.

1.3 Order

ON THE BOARD · IF LANGUAGE · **ORDER**

Same words, different order.

USER READS DAEMON. DAEMON READS USER.

Nothing was added. Nothing was taken away.

Everything changed.

Syntax is the system of rules by which words combine into phrases and sentences. The board's two sentences contain exactly the same three words, yet they describe opposite events. What differs is which noun fills which **thematic role**: the **agent**, who performs the action, and the **patient**, who undergoes it. In English, the default order of a simple sentence is subject, verb, object, and the position of a noun before or after the verb is the main signal of its role.

Not every language relies on order to the same degree. Latin marks roles with endings on the nouns themselves, a system called **case**. *Canis hominem mordet* and *Hominem canis mordet* both mean "the dog bites the man", because the ending *-em* marks *hominem* as the object wherever it stands. English once had a rich case system too, and as it lost most of its endings during the medieval period, word order took over their work.

Order and meaning are separable, and Noam Chomsky (1957) made the point with a famous pair. *Colorless green ideas sleep furiously* is nonsense, yet it is recognisably a grammatical English sentence. *Furiously sleep ideas green colorless*, the same words reversed, is not. Speakers can judge structure independently of meaning, which suggests that the rules of order are a distinct part of what they know.

Order also explains a characteristic stumble in reading. In a **garden-path sentence**, the early words invite one structure and the later words require another. Thomas Bever's (1970) example is *The horse raced past the barn fell.* Most readers first take *raced* as the main verb, reach *fell*, and have to go back: the sentence means "the horse that was raced past the barn fell." Nothing was added and nothing was taken away. The reader had simply assigned the words the wrong structure, and repairing it takes measurable time.

1.4 Ambiguity

ON THE BOARD · IF LANGUAGE · **AMBIGUITY**

One word can carry several meanings.

A BOX holds a daemon. A BOX is where you tick a YES.

The words around it decide which one you meant.

Lexical ambiguity occurs when a single word form carries more than one meaning. Linguists distinguish **homonymy**, in which the meanings are unrelated and only share a form (the *bank* of a river and the *bank* that holds money), from **polysemy**, in which the meanings are related senses of one word (a *paper* as a material and a *paper* as a newspaper). The board's *box*, a container in one sentence and a square to be ticked in the other, is a case of polysemy: both senses descend from the idea of an enclosed space.

How does a listener arrive at the intended meaning? A natural guess is that context is consulted first and only the appropriate meaning retrieved. The evidence suggests something more surprising.

> **RESEARCH SPOTLIGHT · BOTH MEANINGS, THEN ONE**
>
> David Swinney (1979) used a technique called **cross-modal priming**. Participants listened to sentences such as one about a room in which "spiders, roaches and other bugs" were found, a context that clearly favours the insect meaning of *bugs*. At the very moment *bugs* was heard, a string of letters appeared on a screen, and participants decided as quickly as possible whether it was a word. They recognised *ant* faster than an unrelated word, as expected; but they also recognised *spy* faster, although the sentence had nothing to do with spying. When the letters appeared a few syllables later, only the contextually appropriate meaning was still primed. For a brief moment, both meanings had been active, and context then selected one.

Swinney's results, and similar findings by others, suggest that word recognition first makes several meanings available and that context then selects among them, typically within a fraction of a second. Later work showed that the balance depends on how frequent each meaning is and how strongly the context constrains it. The board's last line is therefore correct in a precise sense: the words around a word do decide which meaning was intended, but they do so by selecting among candidates the mind has briefly considered.

Ambiguity is not limited to single words. In **structural ambiguity**, the words are clear but their grouping is not: *I saw the man with the telescope* can mean that the telescope was used for seeing or that the man was holding it. Here too, context and plausibility settle the matter, usually so quickly that the listener never notices there was a choice.

1.5 Spelling

ON THE BOARD · IF LANGUAGE · **SPELLING**

Spelling is a rule for turning a sound into marks.

The rule is old, and full of exceptions.

Nobody is under a spell.

In the corner, in smaller chalk, someone has added: UNDER A SPELL.

It is underlined twice.

A writing system in which letters stand for sounds rests on the **alphabetic principle**. The system of spelling conventions for a particular language is its **orthography**. Orthographies differ in how consistently they map sounds to letters, a property described by the **orthographic depth** hypothesis (Katz & Frost, 1992). In a **shallow orthography**, such as Finnish or Italian, a reader who knows the rules can pronounce nearly any written word correctly. In a **deep orthography**, such as English, the same letters represent different sounds in different words (*through*, *though*, *tough*, *cough*), and the same sound is spelled in different ways.

The board is right that the rule is old and full of exceptions, and the history explains why. English spelling began to settle in the late fifteenth century, after William Caxton set up his printing press at Westminster in 1476, and it was fixed more firmly by the dictionaries of the eighteenth century, most famously Samuel Johnson's of 1755. During the same centuries, the pronunciation of English vowels was changing dramatically in a process called the **Great Vowel Shift**, which ran roughly from 1400 to 1700. Spelling was frozen while the sounds kept moving. Many of the "silent" letters of modern English, such as the *k* of *knight*, were once pronounced.

RESEARCH SPOTLIGHT · WHAT A DEEP ORTHOGRAPHY COSTS

Seymour, Aro and Erskine (2003) compared children learning to read in thirteen European orthographies. Near the end of their first year of instruction, children in most of the languages could read familiar words and simple invented words with high accuracy. Children learning English lagged far behind, and the researchers estimated that establishing basic reading skill in English takes more than twice as long as in the shallowest orthographies. The difference lies not in the children but in the rule they are asked to learn.

Finally, the note in smaller chalk. The verb *spell*, "to name the letters of a word", reached English from Old French *espeller*, which came in turn from a Germanic word meaning to tell or relate. The noun *spell*, an incantation, descends from Old English *spell*, meaning a story or a saying, which survives in *gospel*, "good news". The two words share their distant Germanic ancestry. The connection between writing and magic runs through another word as well: *glamour*, which once meant a magical enchantment, is a Scottish alteration of *grammar*, from an old belief that learning, and the learned, had occult power. Whoever added the chalk note was wrong about the spelling rule, which casts no spell on anyone. They were not entirely wrong about the word.

CHAPTER SUMMARY

- A word is a sign whose form is arbitrarily linked to its meaning; renaming a thing changes the handle, not the thing, although a name can shape what a mind stores about it (1.1).
- Words are built from morphemes, and knowledge of how pieces combine lets speakers understand and produce words they have never encountered (1.2).
- English relies on word order to assign thematic roles; other languages use case endings. Syntax can be judged apart from meaning, and misassigned structure produces garden-path effects (1.3).
- Ambiguous words briefly activate several meanings, and context selects among them within a fraction of a second (1.4).
- English is a deep orthography because its spelling was fixed while its pronunciation was still changing, and this slows children's learning to read (1.5).

KEY TERMS

agent (1.3) · **alphabetic principle** (1.5) · **arbitrariness** (1.1) · **bound morpheme** (1.2) · **case** (1.3) · **cross-modal priming** (1.4) · **deep orthography** (1.5) · **derivational morpheme** (1.2) · **duality of patterning** (intro) · **free morpheme** (1.2) · **garden-path sentence** (1.3) · **Great Vowel Shift** (1.5) · **homonymy** (1.4) · **inflectional morpheme** (1.2) · **lexical ambiguity** (1.4) · **morpheme** (1.2) · **onomatopoeia** (1.1) · **orthographic depth** (1.5) · **patient** (1.3) · **polysemy** (1.4) · **productivity** (intro, 1.2) · **psycholinguistics** (intro) · **reference** (1.1) · **sense** (1.1) · **shallow orthography** (1.5) · **signified** (1.1) · **signifier** (1.1) · **structural ambiguity** (1.4) · **syntax** (1.3) · **thematic role** (1.3)

REVIEW QUESTIONS

1. Explain Saussure's claim that the linguistic sign is arbitrary, and give one apparent exception.
2. Using Frege's distinction, explain why "the morning star is the evening star" was informative to the people who first learned it.
3. What did the wug test show that a vocabulary test could not?
4. Compare how English and Latin mark who did what to whom. Why could English afford to lose most of its case endings?
5. Describe Swinney's cross-modal priming procedure and state what it revealed about the timing of meaning selection.
6. Give two historical reasons that English spelling is irregular, and one consequence for learners.

FURTHER READING

Crystal, D. (2012). *Spell it out: The singular story of English spelling.* Profile Books.

Harley, T. A. (2014). *The psychology of language: From data to theory* (4th ed.). Psychology Press.

Pinker, S. (1994). *The language instinct*. William Morrow.

CHAPTER 2

Attention

THE PLACARD BY THE STAIRS · 2F

What gets looked at, and what that costs everything else.

You cannot keep what you never noticed, so this comes before MEMORY.

The syllabus is on the desk.

Learning Objectives

1. Explain why attention is described as a limited capacity, and describe early theories of selection.
2. Describe the spatial cueing paradigm and the costs and benefits of valid and invalid cues.
3. Distinguish feature search from conjunction search, and explain the pop-out effect.
4. Explain the evidence that apparent multitasking consists of rapid switching between tasks.
5. Describe inattentional blindness and change blindness, and why people underestimate them.

"Everyone knows what attention is," William James wrote in 1890. "It is the taking possession by the mind, in clear and vivid form, of one out of what seem several simultaneously possible objects or trains of thought." The placard gives the reason this chapter precedes the next: you cannot keep what you never noticed. **Attention** is the set of processes by which some information is selected for further processing at the expense of other information, and the phrase "at the expense of" is the heart of the subject.

2.1 Limits

ON THE BOARD · 2F ATTENTION · **LIMITS**

There is only so much of it.

More on one thing is less on everything else.

It is never added. Only moved.

Modern research on attention began with the problem of the **cocktail party**: how can a listener follow one conversation in a crowded room? Colin Cherry (1953) studied it with **dichotic listening**, presenting a different message to each ear through headphones and asking participants to **shadow** one message, repeating it aloud as they heard it. They could do so well, but they could report almost nothing of the message in the other ear. They noticed if the unattended voice changed from male to female, but not if it switched language or was played backwards.

Donald Broadbent (1958) proposed that the mind contains a **filter** that selects one channel early, on the basis of physical features such as location or pitch, and blocks the rest before their meaning is analysed. The theory soon met difficulties. Neville Moray (1959) found that about a third of participants noticed their own name in the unattended ear, which a filter that blocked meaning should not allow. Anne Treisman (1960) proposed instead that the filter **attenuates** unattended input, turning it down rather than off, so that especially important or expected words can still break through.

A different tradition described attention not as a gate but as a resource. In *Attention and Effort*, Daniel Kahneman (1973) proposed a limited pool of **capacity** that can be allocated among tasks, with harder tasks drawing more. This is the board's account: attention is not added, only moved. Devoting more to one thing leaves less for everything else, and the demand of a task, not merely the number of tasks, determines what remains.

2.2 Cues

ON THE BOARD · 2F ATTENTION · **CUES**

A cue says where to look before the thing arrives.

A good cue makes you faster.

A wrong one makes you slower than no cue at all.

Attention can be directed before anything happens. Michael Posner (1980) developed the **spatial cueing paradigm** to measure this. A participant fixes their gaze on the centre of a screen. A cue appears, for example an arrow pointing left or right, and shortly afterwards a target appears on one side, to which the participant responds as quickly as possible. On most trials the cue is **valid**, pointing to where the target will appear; on some it is **invalid**, pointing to the other side; on others a **neutral** cue gives no information.

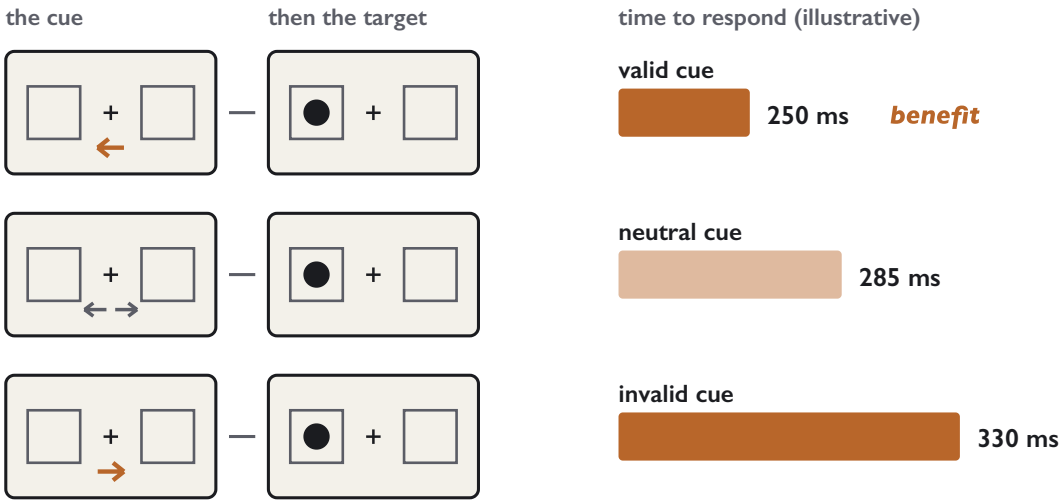


Figure 2.1 The spatial cueing paradigm. A valid cue speeds the response to a target relative to a neutral cue (a benefit); an invalid cue slows it (a cost). Times are illustrative of the pattern, not values from a particular study.

Responses to validly cued targets are faster than after a neutral cue: a **benefit**. Responses to invalidly cued targets are slower than after a neutral cue: a **cost**. The board's claim, that

a wrong cue makes you slower than no cue at all, is precisely Posner's cost. Attention had been moved to the wrong place and had to be withdrawn and redirected.

Cues come in two kinds. **Endogenous** cues, such as an arrow or an instruction, are interpreted and followed voluntarily, and their effect builds over a few hundred milliseconds. **Exogenous** cues, such as a sudden flash at a location, capture attention automatically and quickly, whether or not they predict anything. This is why a flashing light is so hard to ignore even when you know it is irrelevant.

2.3 Search

ON THE BOARD · 2F ATTENTION · **SEARCH**

The more things there are, the longer the one you want takes…

…unless it is the only thing that is different.

The odd one out finds you.

In a **visual search** task, a participant looks for a target among distractors, and the researcher varies the number of items in the display, the **set size**. Anne Treisman and Garry Gelade (1980) found two strikingly different patterns.

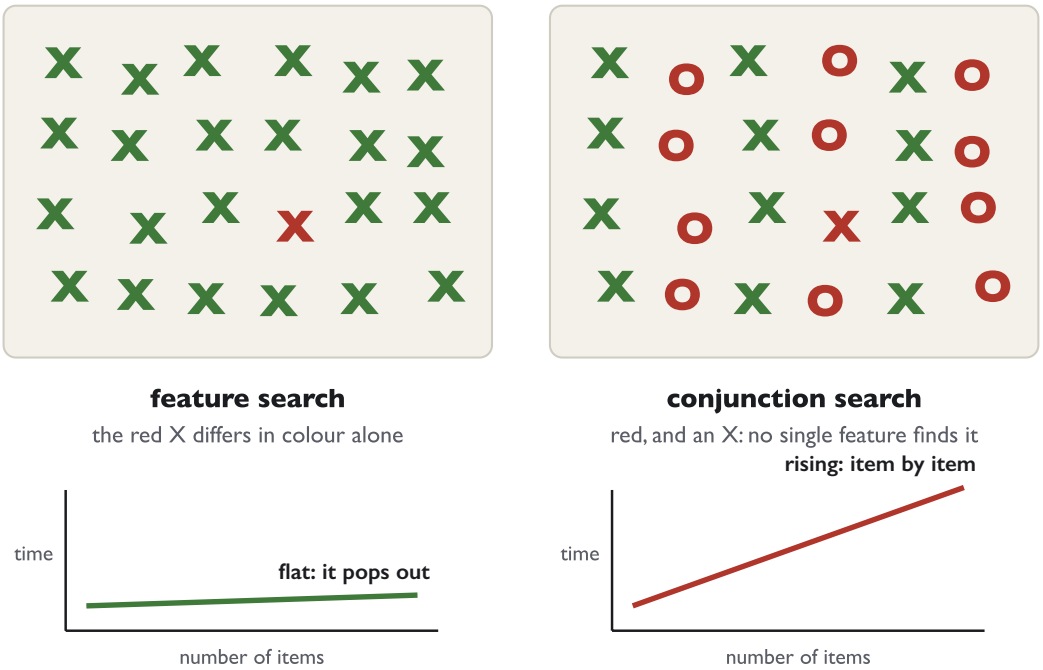


Figure 2.2 Feature search and conjunction search. On the left, the target differs from every distractor in a single feature (colour) and pops out. On the right, it differs only in a combination of features, and must be searched for.

In **feature search**, the target differs from all distractors in a single feature, such as a red item among green ones. Search time barely changes as the set size grows: the target seems to leap out of the display, an effect called **pop-out**. In **conjunction search**, the target is defined by a combination of features, such as a red X among red Os and green Xs. Now search time increases steadily with set size, as if the items were being checked one at a time.

Treisman's **feature integration theory** explains the difference. Simple features such as colour and orientation are registered automatically and in parallel across the whole visual field. Binding features together into a single object requires attention to be directed to that object's location, one location at a time. The board's final line, "the odd one out finds you", describes feature search exactly. Later work, notably Jeremy Wolfe's guided search theory (1994), showed that the two patterns are ends of a continuum: the parallel registration of features is used to guide attention toward likely candidates, so that most real searches lie between the extremes.

2.4 Switching

ON THE BOARD · 2F ATTENTION · **SWITCHING**

Changing what you attend to costs time, every time.

Two things at once is nearly always one thing, then the other, fast.

As long ago as 1927, Arthur Jersild found that students working through a list of numbers were slower when they had to alternate between two operations (adding to one number, subtracting from the next) than when they performed the same operation throughout. The extra time is the **switch cost**. Rogers and Monsell (1995) showed that it shrinks when people are given time to prepare for the coming switch, but that it does not disappear: some part of reconfiguring for a new task cannot be done in advance.

What looks like doing two things at once is usually something else. In the **psychological refractory period** paradigm, two stimuli arrive in quick succession, each requiring its own response. The closer together they arrive, the more the second response is delayed. Harold Pashler (1994) argued that this reflects a central **bottleneck**: the stage at which a response is selected can serve only one task at a time, so the second task waits.

RESEARCH SPOTLIGHT · TALKING WHILE DRIVING

Strayer and Johnston (2001) asked participants to perform a simulated driving task while conversing on a mobile telephone. During conversation, participants missed more traffic signals and responded more slowly to the signals they did detect. Hands-free telephones produced the same impairment as hand-held ones. Listening to the radio or to a book on tape did not. The difficulty was not in the hands but in the conversation, which competed with driving for the same limited attention.

2.5 Missed

ON THE BOARD · 2F ATTENTION · **MISSED**

Watch one thing closely enough and something unmissable walks past.

Nobody who missed it believes they did.

Inattentional blindness is the failure to notice a fully visible but unexpected object when attention is engaged elsewhere. Arien Mack and Irvin Rock (1998) studied it systematically with brief displays. The most famous illustration came a year later.

RESEARCH SPOTLIGHT · GORILLAS IN OUR MIDST

Daniel Simons and Christopher Chabris (1999) filmed two teams passing basketballs and asked viewers to count the passes made by the team in white. Midway through the video, a person in a gorilla costume walked into the middle of the game, faced the camera, thumped its chest and walked off. Roughly half of the viewers did not notice the gorilla at all. Many who were later shown the video again were astonished, and some insisted that it must have been a different video.

A related failure, **change blindness**, is the failure to notice a change to a scene, especially when the change coincides with an interruption. In a field study by Simons and Levin (1998), an experimenter asked a pedestrian for directions; during the conversation, two people carrying a door passed between them, and the experimenter was replaced by a different person. About half of the pedestrians did not notice that they were now talking to someone else.

The board's last line is supported by research too. People are confident that they would notice such events, and greatly overestimate how often they would; Levin and colleagues (2000) called this misplaced confidence **change blindness blindness**. The finding connects attention to the last floor of the building: being sure that one saw everything is not evidence that one did.

CHAPTER SUMMARY

- Attention is limited. Early filter theories gave way to attenuation and capacity theories; on all of them, more attention to one thing means less for others (2.1).
- A valid cue speeds responses and an invalid cue slows them relative to no cue; voluntary and automatic cues differ in speed and control (2.2).
- A target differing in a single feature pops out regardless of set size; a target defined by a conjunction must be searched for (2.3).
- Switching between tasks has a cost that preparation reduces but does not remove; apparent multitasking is usually rapid alternation through a central bottleneck (2.4).
- Unexpected objects and changes are frequently missed when attention is engaged, and people underestimate how often this happens (2.5).

KEY TERMS

attention (intro) · **attenuation** (2.1) · **benefit** (2.2) · **bottleneck** (2.4) · **capacity** (2.1) · **change blindness** (2.5) · **change blindness blindness** (2.5) · **cocktail party problem** (2.1) · **conjunction search** (2.3) · **cost** (2.2) · **dichotic listening** (2.1) · **endogenous cue** (2.2) · **exogenous cue** (2.2) · **feature integration theory** (2.3) · **feature search** (2.3) · **filter** (2.1) · **inattentional blindness** (2.5) · **invalid cue** (2.2) · **pop-out** (2.3) · **psychological refractory period** (2.4) · **set size** (2.3) · **shadowing** (2.1) · **spatial cueing paradigm** (2.2) · **switch cost** (2.4) · **valid cue** (2.2) · **visual search** (2.3)

REVIEW QUESTIONS

1. What did Moray's own-name finding imply for Broadbent's filter theory, and how did Treisman revise the theory?
2. In Posner's paradigm, why is a neutral cue needed to separate costs from benefits?

3. Sketch the relation between set size and search time for feature search and for conjunction search, and explain the difference using feature integration theory.
4. Why did Strayer and Johnston's hands-free condition matter for the interpretation of their results?
5. Distinguish inattentional blindness from change blindness, with one study of each.

FURTHER READING

Chabris, C., & Simons, D. (2010). *The invisible gorilla: And other ways our intuitions deceive us.* Crown.

Pashler, H. E. (1998). *The psychology of attention.* MIT Press.

CHAPTER 3

Memory

THE PLACARD BY THE STAIRS · 3F

What is kept, what fades, and what changes each time you take it out.

The syllabus is on the desk.

Learning Objectives

1. Describe memory as encoding, storage and retrieval, and explain the evidence that memories are reconstructed.
2. Describe the shape of the forgetting curve and the evidence that forgetting is never quite complete.
3. Distinguish proactive from retroactive interference and explain the role of similarity.
4. Explain how recall can alter a memory, with evidence from misinformation and reconsolidation research.
5. State the spacing effect and the testing effect, and explain why learners tend to doubt them.

Psychologists divide remembering into three stages: **encoding**, the process by which information is taken in and transformed into a form that can be kept; **storage**, the keeping of it over time; and **retrieval**, bringing it back when it is needed. A failure of memory can occur at any of the three. This chapter is concerned mainly with **long-term memory**, the durable store of facts, events and skills, and its central lesson is the one the placard announces: memory changes each time it is taken out.

3.1 Storing

ON THE BOARD · 3F MEMORY · **STORING**

Nothing is kept whole.

What is kept is enough to build it again.

It is tempting to imagine memory as a recording that can be replayed. The evidence points elsewhere. Frederic Bartlett (1932) asked English participants to read a Native American folk tale, *The War of the Ghosts*, and to reproduce it from memory, repeatedly, over intervals of minutes to years. Their versions grew shorter, lost details that seemed strange to them, and were reshaped to fit their own expectations: unfamiliar elements were replaced with familiar ones, and puzzling events were given explanations the original did not contain. Bartlett concluded that remembering is **reconstructive**. What is stored is not a copy but an organised outline, which he called a **schema**, together with a few details, and the memory is rebuilt from them at recall.

The gist survives better than the surface. Jacqueline Sachs (1967) had people listen to passages and then judge whether a test sentence had appeared in them. After a short delay, people readily noticed a change in meaning but often failed to notice a change in wording that preserved the meaning. The exact words had been let go; the meaning had been kept.

What gets kept depends on how it was encoded. Craik and Lockhart's (1972) **levels of processing** framework proposed that information processed for its meaning (deep processing) is better remembered than information processed for its appearance or sound (shallow processing). In a classic test, Craik and Tulving (1975) found that words judged for whether they fitted a sentence were remembered far better than words judged for whether they were printed in capital letters. Storing, in the board's sense, is building the materials from which a memory can later be built again; the deeper the building, the better the later reconstruction.

3.2 Fading

ON THE BOARD · 3F MEMORY · **FADING**

Left alone, it fades.

Fastest at first. Then slower. Never quite to nothing.

The first rigorous experiments on memory were performed by Hermann Ebbinghaus on himself, and published in 1885. To avoid the influence of meaning, he memorised lists of **nonsense syllables** such as *DAX* and *BUP*, and measured forgetting by the **savings** method: how much less time it took to relearn a list to perfection than it had taken to learn it the first time. Savings could reveal memory even when he could recall nothing.

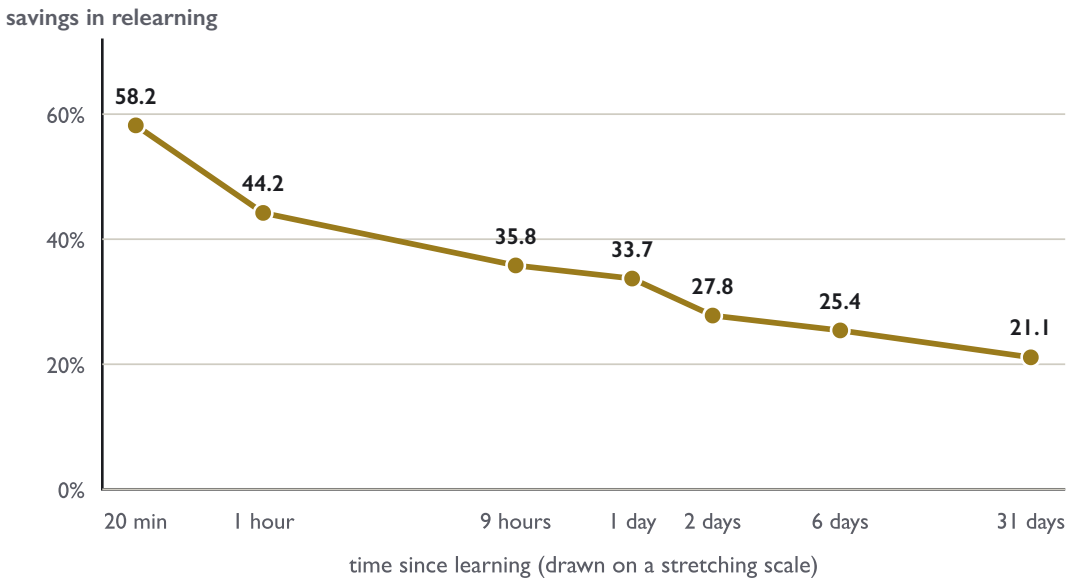


Figure 3.1 Ebbinghaus's forgetting curve: the savings in relearning time after delays from twenty minutes to a month. Most of the loss occurs in the first hours; after that, it slows and levels off.

The resulting **forgetting curve** has exactly the shape the board describes. The loss is steepest at first, more than half of the savings disappearing within a day, and then becomes slower and slower. Murre and Dros (2015) repeated the experiment more than a century later and obtained a closely similar curve. Mathematically, forgetting is better de-

scribed by a power function than by a simple exponential decay (Wixted & Ebbesen, 1991), which captures the continuing slowdown.

Forgetting also rarely reaches zero. Harry Bahrick (1984) tested people who had studied Spanish at school between one and fifty years earlier. Their knowledge declined over the first three to six years and then remained remarkably stable for decades before a late decline, a durable residue he called the **permastore**. Never quite to nothing is a fair summary of the evidence.

3.3 Crowding

ON THE BOARD · 3F MEMORY · **CROWDING**

New things push on old ones, and old on new.

The most alike are the most easily confused.

Is forgetting simply fading with time, or are memories pushed out by other memories? John McGeoch (1932) argued for the second view: time itself does nothing, any more than time rusts iron; it is what happens during the time that matters. The best-known support came from Jenkins and Dallenbach (1924), whose participants recalled more after a period of sleep than after an equal period awake, when new experiences could intervene.

The pushing runs in both directions, as the board says. In **retroactive interference**, new learning impairs memory for older material: after learning a second list, recall of the first suffers. In **proactive interference**, old learning impairs memory for newer material. Benton Underwood (1957) showed how powerful proactive interference can be. Reviewing many studies, he found that the amount a participant forgot of a list over a day depended heavily on how many lists that participant had learned before it in the laboratory. Much of what had looked like forgetting over time was interference from earlier learning.

Interference depends on similarity. Material that resembles what was learned before interferes most, because the cues that should retrieve one item now retrieve its rivals as well. Wickens, Born and Allen (1963) showed the converse in short-term memory: over a run of brief recall trials with items of one kind, such as consonants, recall fell from trial to tri-

al; when the items switched to a different kind, such as digits, recall recovered sharply, a **release from proactive interference**. The most alike are the most easily confused; a change of kind brings relief.

> **TRY IT**
>
> Learn the telephone extension of a new colleague, then of another, then of a third. Now recall the first. Next, learn a fourth number that belongs to something quite different, such as a locker combination, and notice how much more easily it stays separate. The first set competed; the last did not.

3.4 Recall

ON THE BOARD · 3F MEMORY · **RECALL**

Remembering is not reading a copy.

It is building one, and the build is saved over the old.

Remember it often enough, and you remember the remembering.

If memory is reconstructed, every act of recall is an opportunity for it to change. Elizabeth Loftus has shown how easily this happens.

> **RESEARCH SPOTLIGHT · SMASHED OR HIT?**
>
> Loftus and Palmer (1974) showed participants films of car accidents and asked how fast the cars were going when they "hit" each other, or when they "smashed into", "collided with", "bumped" or "contacted" each other. The verb changed the estimates: "smashed" produced the highest speeds, "contacted" the lowest. A week later, participants who had been asked the "smashed" question were more likely to report having seen broken glass, although there had been none in the film. A single word in a question had become part of the memory of the event.

This **misinformation effect** shows that information encountered after an event can be incorporated into the memory of the event itself. The board describes the mechanism: recall

builds a version of the memory, and the version is saved over the old.

Neuroscience has found a biological counterpart. A memory that has been consolidated, made stable, was long thought to be fixed. Nader, Schafe and LeDoux (2000) showed that in rats, reactivating a consolidated memory returned it to a fragile state, in which blocking protein synthesis in the brain could disrupt it. Retrieved memories, they argued, must be stored again, a process called **reconsolidation**, and during that process they can be modified.

The board's last line is borne out by studies of vivid memories of public events. Talarico and Rubin (2003) asked students about the September 11 attacks the day after they occurred, and again weeks or months later. The consistency of their memories declined over time, just as the consistency of their memories for ordinary events did, but their confidence in the attack memories remained high. With repeated retelling, what people remember is increasingly their previous rememberings.

A CLOSER LOOK · HOW FAR DOES IT GO?

Research on altered and false memories has had real consequences in courtrooms and therapy, and has also been debated. It is well established that details of real memories can be altered by later information. Whether entire events can be implanted, and how often, is more contested: studies such as Loftus and Pickrell's (1995) "lost in the mall" found that a minority of participants came to report partial memories of a suggested childhood event, and later reviews disagree about how such reports should be counted. The safe conclusion is the board's: recall is a building, not a reading, and what is built can differ from what was there.

3.5 Spacing

ON THE BOARD · 3F MEMORY · **SPACING**

Once a day for seven days beats seven times in one day.

Nobody believes this until they try it.

The **spacing effect** is the finding that study distributed over time produces better long-term retention than the same amount of study massed together. Ebbinghaus noticed it in his own learning, and it has since been confirmed in hundreds of experiments. A meta-analysis by Cepeda and colleagues (2006) reviewed over three hundred experiments and found that distributed practice reliably outperformed massed practice, and that the optimal gap between study sessions grows with the length of time over which the material must be kept.

A second, related finding is the **testing effect**, or retrieval practice effect: practising recall strengthens memory more than re-reading does. Roediger and Karpicke (2006) had students study a prose passage and then either study it again or take a recall test on it. Tested after five minutes, the re-studiers did slightly better. Tested after a week, the students who had practised recall did substantially better.

The board's last sentence may be its most important. Learners consistently misjudge these effects. Massed study feels effective because the material is fluent and familiar at the end of the session, and that fluency is mistaken for learning. Kornell (2009) had students learn vocabulary with flashcards in spaced or massed schedules. Spacing was more effective for the great majority of participants, yet most of them believed that massing had served them better. Bjork (1994) called conditions such as spacing **desirable difficulties**: they make learning feel harder while making it last longer.

> **TRY IT**
>
> Take two short lists of unfamiliar words. Study the first seven times in one sitting. Study the second once a day for seven days. A week after the last session, test yourself on both. Most people predict the first list will win.

CHAPTER SUMMARY

- Memory is reconstructive: a schema and some details are stored, meaning is kept better than wording, and deeper encoding yields better recall (3.1).
- Forgetting is fastest at first and slows continuously, and some knowledge persists for decades (3.2).

- Interference, both proactive and retroactive, accounts for much forgetting, and is greatest between similar materials (3.3).
- Recall rebuilds a memory, which can incorporate later information; retrieved memories are restabilised by reconsolidation, and confidence can stay high as accuracy falls (3.4).
- Spaced study and retrieval practice produce durable learning, although learners tend to believe the opposite (3.5).

KEY TERMS

desirable difficulties (3.5) · **encoding** (intro) · **forgetting curve** (3.2) · **levels of processing** (3.1) · **long-term memory** (intro) · **misinformation effect** (3.4) · **nonsense syllables** (3.2) · **permastore** (3.2) · **proactive interference** (3.3) · **reconsolidation** (3.4) · **reconstructive memory** (3.1) · **release from proactive interference** (3.3) · **retrieval** (intro) · **retroactive interference** (3.3) · **savings** (3.2) · **schema** (3.1) · **spacing effect** (3.5) · **storage** (intro) · **testing effect** (3.5)

REVIEW QUESTIONS

1. What changes did Bartlett observe in repeated reproductions of *The War of the Ghosts*, and what did he conclude?
2. Why did Ebbinghaus measure savings rather than recall alone?
3. Describe one result that favours interference over decay as a cause of forgetting.
4. How did Loftus and Palmer show that a question can alter a memory?
5. Why do learners prefer massed study, and why are they mistaken?

FURTHER READING

Brown, P. C., Roediger, H. L., & McDaniel, M. A. (2014). *Make it stick: The science of successful learning*. Belknap Press.

Schacter, D. L. (2001). *The seven sins of memory: How the mind forgets and remembers*. Houghton Mifflin.

CHAPTER 4

Categories

THE PLACARD BY THE STAIRS · 4F

How things are sorted, and why the edges will not hold still.

The syllabus is on the desk.

Learning Objectives

1. Contrast the classical view of categories with prototype and exemplar views.
2. Describe evidence for typicality effects and graded category membership.
3. Explain why the features chosen for sorting determine which things are grouped together.
4. Describe the basic level of categorisation and how expertise shifts it.
5. Explain the role and the warning signs of residual categories.

To **categorise** is to treat different things as equivalent for some purpose: to recognise this animal as a dog, this object as a chair, this situation as an emergency. Categories let us generalise from what we know about some members to members we have never met. For two thousand years, the dominant account was the **classical view**, traceable to Aristotle: a category is defined by a set of **necessary and sufficient conditions**. Every member has all the defining features, and anything with all of them is a member. The research of the twentieth century showed that most everyday categories do not work this way.

The first doubt came from philosophy. Ludwig Wittgenstein, in *Philosophical Investigations* (1953), asked what all games have in common. Some involve winning and losing, some do not; some involve skill, some luck; some are played alone. He concluded that there is no single shared feature, only "a complicated network of similarities overlapping and criss-crossing", which he called **family resemblance**.

4.1 Middles

ON THE BOARD · 4F CATEGORIES · **MIDDLES**

Most kinds have a middle: the best example, the one that comes first.

Everything else is more or less like it.

Eleanor Rosch (1975) asked people to rate how good an example each member of a category was. Agreement was high. A robin was rated an excellent example of a bird; a chicken less so; a penguin or an ostrich a poor one. These **typicality** ratings predicted performance in many tasks. People verify that a robin is a bird faster than that a penguin is a bird (Rips, Shoben & Smith, 1973); asked to list members of a category, they name typical members first and most often; and children learn typical members of categories earlier.

Prototype theory explains these effects by proposing that a category is represented by its **prototype**, a summary of the category's central tendency, and that an item is categorised by its similarity to the prototype. An alternative, **exemplar theory** (Medin & Schaffer, 1978), proposes that people store many individual examples and categorise a new item by its similarity to the stored examples. Both explain the board's observation that a category has a middle and that everything else is more or less like it. They differ in what the middle is: a summary on one account, a dense cluster of remembered cases on the other.

4.2 Edges

ON THE BOARD · 4F CATEGORIES · **EDGES**

No kind has a clean edge.

Somewhere there is always a thing that is half in.

If categories have middles, what happens at their boundaries? William Labov (1973) showed people line drawings of containers that varied gradually in the ratio of width to depth and asked them to name each one. Narrow containers were called cups, wide ones bowls, but there was no sharp line between them, and the boundary moved with context.

When participants imagined the container holding coffee, more drawings were called cups; when they imagined it holding mashed potatoes, more were called bowls.

RESEARCH SPOTLIGHT · UNSURE, AND INCONSISTENT

McCloskey and Glucksberg (1978) asked people whether each of a long list of items belonged to categories such as furniture or fruit. For clear members and clear non-members, people agreed with one another and with themselves. For borderline items (is a bookend furniture?), they disagreed with one another, and when asked again a month later, many changed their own answers. The edges of everyday categories are not merely unknown; they are not fixed.

The law has had to draw such lines by decree. In *Nix v. Hedden* (1893), the Supreme Court of the United States had to decide whether tomatoes were fruits or vegetables for the purposes of a tariff on vegetables. It acknowledged that botanically a tomato is a fruit, and ruled that in common speech, and therefore for the tariff, it is a vegetable. A category used for one purpose drew its edge in one place; a category used for another drew it elsewhere.

4.3 Features

ON THE BOARD · 4F CATEGORIES · **FEATURES**

Sorting looks at a handful of features.

Pick the wrong ones, and two very different things share a drawer.

Sorting depends on which features are considered. A sorter attending to wings will put birds, bats and butterflies together; a sorter attending to how the young are fed will put bats with mice and whales. Neither is wrong as a sorting; they serve different purposes. Whales were long classed with fish on the basis of their shape and habitat, and it was only when naturalists such as Carl Linnaeus attended to other features, warm blood, lungs, and the suckling of young, that they were moved to the mammals in his *Systema Naturae* of 1758.

Rosch and Mervis (1975) showed that the features of natural categories are not randomly distributed. Typical members share many features with other members of their category and few with members of contrasting categories; that is what makes them typical. Features also come in correlated bundles: things with feathers tend to have wings and beaks, and the mind exploits such correlations.

Similarity alone, however, cannot be the whole story, because almost any two things are similar in some respect. Murphy and Medin (1985) argued that categories are held together by people's **theories**, their background knowledge of why features go together. A person who sees someone at a party jump fully clothed into a swimming pool may categorise them as drunk, not because jumping into pools is a feature of drunkenness but because their theory of drunkenness explains the act. The features that matter are the ones a theory makes relevant. Pick the wrong ones, as the board warns, and very different things share a drawer.

4.4 Levels

ON THE BOARD · 4F CATEGORIES · **LEVELS**

The same thing sits in several kinds at once, at different heights.

A thing. A kind of thing. A kind of kind.

The same object can be called a kitchen chair, a chair, a piece of furniture or a thing. These names form a **taxonomy**, a hierarchy in which each category includes those below it. Rosch and her colleagues (1976) found that one level is psychologically privileged. At the **basic level** (chair, dog, car), categories are most informative for the least effort: their members share many features with one another and differ clearly from the members of neighbouring categories. Above them, **superordinate** categories (furniture, animal, vehicle) have members that share few features; below them, **subordinate** categories (kitchen chair, beagle, saloon car) differ only slightly from their neighbours.

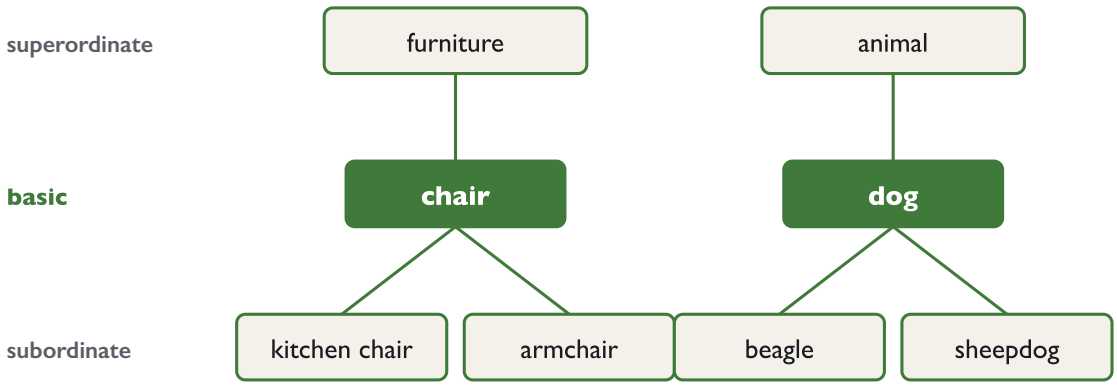


Figure 4.1 Three levels of one taxonomy. The basic level, in the middle, is the level at which objects are named most readily, learned first by children and recognised fastest.

The basic level is the one at which people name objects spontaneously, at which children first learn names, and at which pictures are identified fastest. Hierarchies also show up in timing. Collins and Quillian (1969) found that people take longer to verify that a canary is an animal than that a canary is a bird, as if travelling up a hierarchy of categories one level at a time, although later work showed that typicality and familiarity explain much of that pattern too.

The basic level is not fixed. Tanaka and Taylor (1991) found that experienced bird-watchers and dog experts identified pictures in their area of expertise as quickly at the subordinate level (a robin, a beagle) as at the basic level. Expertise moves the privileged level down the hierarchy.

4.5 Other

ON THE BOARD · 4F CATEGORIES · **OTHER**

Every way of sorting ends up with a drawer called OTHER.

Watch how fast it fills.

No categorisation scheme fits the world perfectly, and every practical scheme therefore ends with a **residual category**: *other, miscellaneous, not elsewhere classified*. In their study of classification systems, Bowker and Star (1999) examined, among others, the International Classification of Diseases, and showed how residual categories are both indispensable and revealing. They are where the scheme admits that its categories did not anticipate everything.

Jorge Luis Borges satirised the arbitrariness of classification in a 1942 essay describing an imaginary Chinese encyclopedia that divides animals into, among other kinds, those belonging to the Emperor, those drawn with a very fine camel-hair brush, those that have just broken the flower vase, and *et cetera*. The philosopher Michel Foucault (1966) said that the passage made him laugh, and then made him think about how every culture's order of things is only one order among many.

The board's warning is practical. A residual category that grows quickly is information. It signals that the world contains a kind the scheme was not built to recognise, or that the features chosen for sorting are no longer the right ones. Classification systems are revised, in libraries, in medicine and in biology, most often when their *other* drawer has grown too large to ignore.

CHAPTER SUMMARY

- Most everyday categories are not defined by necessary and sufficient conditions; they have members that are more and less typical, explained by prototype and exemplar theories (intro, 4.1).
- Category boundaries are graded, shift with context, and are inconsistent even within one person (4.2).
- Which features are attended to determines what is grouped; theories make some features relevant (4.3).
- Categories form hierarchies with a privileged basic level, which expertise shifts downward (4.4).
- Every practical scheme has a residual category, and its growth signals that the scheme needs revision (4.5).

KEY TERMS

basic level (4.4) · **categorise** (intro) · **classical view** (intro) · **exemplar theory** (4.1) · **family resemblance** (intro, 4.3) · **necessary and sufficient conditions** (intro) · **prototype** (4.1) · **prototype theory** (4.1) · **residual category** (4.5) · **subordinate level** (4.4) · **superordinate level** (4.4) · **taxonomy** (4.4) · **theory-based categorisation** (4.3) · **typicality** (4.1)

REVIEW QUESTIONS

1. State the classical view of categories and describe Wittgenstein's objection to it.
2. Give three findings that show typicality matters to how categories are used.
3. How did Labov's cups and bowls study show that category boundaries depend on context?
4. Why can similarity alone not explain categorisation? Use Murphy and Medin's example.
5. What makes the basic level basic, and how does expertise change it?
6. Why might the growth of a residual category be a useful warning?

FURTHER READING

Bowker, G. C., & Star, S. L. (1999). *Sorting things out: Classification and its consequences*. MIT Press.

Murphy, G. L. (2002). *The big book of concepts*. MIT Press.

CHAPTER 5

Learning

THE PLACARD BY THE STAIRS · 5F

How a mind changes itself, one small join at a time.

The syllabus is on the desk.

Learning Objectives

1. Describe the neuron doctrine and the synapse, and explain what is meant by the strength of a synaptic connection.
2. State Hebb's postulate and describe long-term potentiation as evidence for it.
3. Explain the firing threshold, summation, and why cells differ in how readily they fire.
4. Explain why feedback is necessary for skill, with evidence from classic and modern studies.
5. Describe synaptic pruning and sensitive periods in development.

In everyday speech, **learning** means acquiring knowledge or skill. Psychologists define it more broadly, as a relatively lasting change in behaviour or knowledge that results from experience. The placard adds the question this chapter answers: if a mind changes itself, what physically changes? The answer, established over the last century and a half, is that experience alters the connections between the brain's cells, one small join at a time.

5.1 Connections

ON THE BOARD · 5F LEARNING · **CONNECTIONS**

A mind is cells joined to cells.

Each join has a strength: how hard one cell pushes the next when it fires.

In the late nineteenth century, it was disputed whether the nervous system was a single continuous web or was made of separate cells. Using a staining method invented by Camillo Golgi, the Spanish anatomist Santiago Ramón y Cajal showed that it is made of discrete cells, the **neurons**, which touch without fusing. This is the **neuron doctrine**. Golgi and Cajal shared the 1906 Nobel Prize for their work on the structure of the nervous system, although Golgi used his Nobel lecture to reject the doctrine. In 1897 the physiologist Charles Sherrington gave the point of contact between two neurons its name, the **synapse**.

A typical neuron receives signals through branching **dendrites** and sends its own signal along a single **axon**, whose terminals form synapses on other cells. The human brain contains roughly eighty-six billion neurons (Azevedo et al., 2009), and a single neuron may form thousands of synapses. When a neuron fires, its terminals release chemical messengers, **neurotransmitters**, that alter the electrical state of the receiving cell. At an **excitatory synapse** the message pushes the receiving cell toward firing; at an **inhibitory synapse** it holds the cell back. In the brain's outer layer, glutamate is the main excitatory transmitter and GABA the main inhibitory one.

What the board calls the strength of a join, physiologists call **synaptic strength** or synaptic efficacy: how large a change one firing of the sending cell produces in the receiving cell. Strengths differ from synapse to synapse, and, crucially, they change.

5.2 Practice

ON THE BOARD · 5F LEARNING · **PRACTICE**

Use a join and it grows stronger.

Cells that fire together wire together, as the saying goes.

In *The Organization of Behavior* (1949), the Canadian psychologist Donald Hebb proposed that when one cell repeatedly takes part in firing another, some growth process or metabolic change takes place in one or both cells, so that the first becomes more efficient at firing the second. Groups of cells strengthened in this way would come to act together as **cell assemblies**, the physical basis, Hebb suggested, of an idea. The proposal is known as **Hebb's postulate**, or Hebbian learning. Its popular summary, "cells that fire together wire together", is usually credited to the neuroscientist Carla Shatz (1992), who used it to describe how connections are refined in the developing visual system.

RESEARCH SPOTLIGHT · LONG-TERM POTENTIATION

Hebb's postulate waited a quarter of a century for direct physiological evidence. Bliss and Lømo (1973) stimulated a pathway into the hippocampus of anaesthetised rabbits with brief, high-frequency bursts. Afterwards, single test pulses along the same pathway produced a larger response in the receiving cells than before, and the increase lasted for hours. This **long-term potentiation** (LTP), a lasting increase in synaptic strength produced by intense activity, became the most studied cellular mechanism of learning and memory.

Work on simpler animals showed the same principle at the level of single synapses. Eric Kandel's studies of the sea slug *Aplysia*, for which he shared the Nobel Prize in 2000, traced simple forms of learning to changes in the strength of identified synapses, and showed that lasting memories require the synthesis of new proteins and the growth of new synaptic connections.

Practice changes structure on a larger scale too. Maguire and colleagues (2000) found that licensed London taxi drivers, who spend years learning the city's streets, had a larger posterior hippocampus than comparison participants, and the size was related to their years of driving. At the level of behaviour, Fitts and Posner (1967) described skill acquisition in

three stages: a **cognitive stage**, in which the learner thinks through each step; an **associative stage**, in which errors fall and the steps join together; and an **autonomous stage**, in which the skill runs with little attention.

5.3 Threshold

ON THE BOARD · 5F LEARNING · **THRESHOLD**

A cell fires only when the pushes on it add up past a line.

Some cells sit close to that line. They fire at almost nothing.

A neuron at rest holds an electrical charge across its membrane, its **resting potential**, of about -70 millivolts (the inside negative relative to the outside). Signals arriving at excitatory synapses push this potential upward; signals at inhibitory synapses push it down. If the potential at the start of the axon rises past a **threshold**, typically around -55 millivolts, the cell fires an **action potential**: a brief, self-propagating reversal of charge that travels down the axon. The action potential is **all-or-none**. A stronger push does not produce a bigger action potential; it produces one, or several in quicker succession, or none.

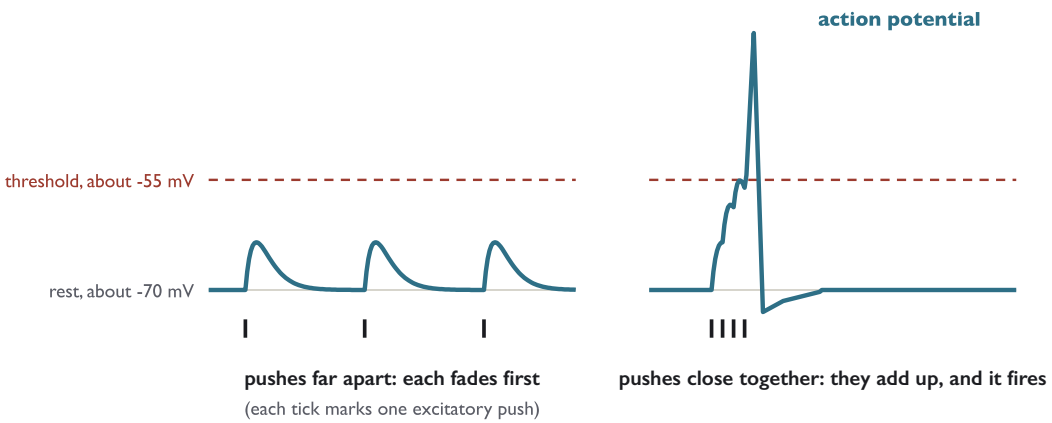


Figure 5.1 Summation and the threshold. Single excitatory pushes (left) raise the membrane potential but fall short. Pushes arriving close together in time add up (right), cross the threshold, and the cell fires.

Because a single synapse rarely moves the cell far enough by itself, the cell's decision depends on **summation**. In **temporal summation**, several signals arriving at one synapse in quick succession add up before the first has faded. In **spatial summation**, signals arriving at different synapses at the same moment add up. Inhibitory signals subtract. The cell, in effect, adds its pushes and fires only when the sum crosses the line, exactly as the board says. The ionic basis of the action potential was worked out by Alan Hodgkin and Andrew Huxley (1952) in the giant axon of the squid, work for which they shared the Nobel Prize in 1963.

Cells differ in how close to the line they sit. A neuron whose resting potential lies nearer its threshold, or whose threshold is lower, will fire in response to small inputs that leave its neighbours silent. This **excitability** is not fixed. It is adjusted by **neuromodulators** such as acetylcholine and noradrenaline, which change how readily whole populations of cells respond, and it can itself change with experience, a process called **intrinsic plasticity** (Zhang & Linden, 2003). Learning, then, can alter not only how hard one cell pushes another but how easily each cell is moved at all.

5.4 Feedback

ON THE BOARD · 5F LEARNING · **FEEDBACK**

Learning needs to find out how it went.

Without that, practice only makes a habit, not a skill.

Edward Thorndike (1898) placed hungry cats in **puzzle boxes** from which they could escape, and reach food, only by operating a latch. Over many trials, the movements that opened the box became more frequent and the useless ones dropped away. Thorndike summarised the result in the **law of effect**: responses followed by a satisfying outcome become more likely to recur, and those followed by an unsatisfying one less likely. The outcome is information, and without it there is nothing to select.

RESEARCH SPOTLIGHT · LINES DRAWN BLINDFOLDED

In a series of experiments reported in *Human Learning* (1931), Thorndike had blindfolded participants draw lines intended to be a set length, thousands of times over several days, without being told how close they came. Their accuracy did not improve. Trowbridge and Cason (1932) repeated the task with groups that were told the size of their error after each attempt, and those groups improved steadily. Repetition alone had produced consistency, a habit, but not accuracy. Only practice informed by its results produced skill.

Modern research on expertise reaches the same conclusion. Ericsson, Krampe and Tesch-Römer (1993) found that the best violinists at a music academy had accumulated far more hours of **deliberate practice** than less accomplished students: practice designed to improve specific weaknesses, with immediate feedback and opportunities to correct. Merely playing, however often, did not have the same effect.

A CLOSER LOOK · NOT ALL FEEDBACK HELPS

It does not follow that any feedback improves performance. In a large meta-analysis, Kluger and DeNisi (1996) found that feedback interventions improved performance on average, but that in over a third of cases they made performance worse. Feedback directed at the task, and at how to do it better, tended to help; feedback that drew attention to the person, such as praise or criticism of the learner, often did not. Learning needs to find out how it went, but what it finds out must be about the work.

5.5 Pruning

ON THE BOARD · 5F LEARNING · **PRUNING**

Joins that are never used are taken away.

Growing up is losing most of them, on purpose.

The developing brain overproduces connections and then removes many of them. Peter Huttenlocher (1979) counted synapses in human frontal cortex at different ages and found that synaptic density rises rapidly after birth, peaks in early childhood, and then de-

clines through later childhood and adolescence toward adult levels. Huttenlocher and Dabholkar (1997) found that the timing differs across the brain: synaptic density in the visual cortex peaks within the first year, while in the prefrontal cortex the decline continues into adolescence. This **synaptic pruning** is selective. Connections that are used are stabilised; those that are not are eliminated.

Pruning makes development sensitive to experience during particular windows, known as **sensitive periods** (or, when the window closes sharply, critical periods). David Hubel and Torsten Wiesel showed in the 1960s that if one eye of a kitten was kept closed during an early period of life, the connections from that eye to the visual cortex lost out to the connections from the open eye, and the effect persisted; the same closure in an adult cat had little effect. They shared the Nobel Prize in 1981.

Human perception shows the same pattern. Werker and Tees (1984) found that infants of six to eight months could distinguish speech sounds used in languages they had never heard, such as contrasts from Hindi and from a language of the Pacific Northwest, while infants of ten to twelve months raised in English-speaking homes largely could not. The older infants had not failed to learn; they had specialised in the sounds their own language uses.

The board compresses this into a striking sentence, and it deserves one qualification. Growing up does mean losing connections on purpose, and in some regions synaptic density falls by something like a third to a half between early childhood and adulthood. Whether that amounts to "most" depends on the region and the measure. The principle is not in doubt: a developing brain is shaped as much by what it removes as by what it adds.

CHAPTER SUMMARY

- The nervous system is made of discrete neurons joined at synapses, each with a strength that can change (5.1).
- Connections strengthen with correlated activity, as Hebb proposed; long-term potentiation is a cellular mechanism of this strengthening, and skill passes through cognitive, associative and autonomous stages (5.2).
- A neuron fires only when summed excitatory and inhibitory inputs cross its threshold; excitability differs between cells and can itself change (5.3).

- Feedback is necessary for skill; practice without knowledge of results produces consistency but not accuracy, and feedback helps most when it is about the task (5.4).
- Development removes unused synapses, which makes learning especially sensitive to experience during certain periods (5.5).

KEY TERMS

action potential (5.3) · **all-or-none** (5.3) · **associative stage** (5.2) · **autonomous stage** (5.2) · **axon** (5.1) · **cell assembly** (5.2) · **cognitive stage** (5.2) · **deliberate practice** (5.4) · **dendrite** (5.1) · **excitability** (5.3) · **excitatory synapse** (5.1) · **Hebb's postulate** (5.2) · **inhibitory synapse** (5.1) · **intrinsic plasticity** (5.3) · **law of effect** (5.4) · **learning** (intro) · **long-term potentiation** (5.2) · **neuromodulator** (5.3) · **neuron** (5.1) · **neuron doctrine** (5.1) · **neurotransmitter** (5.1) · **puzzle box** (5.4) · **resting potential** (5.3) · **sensitive period** (5.5) · **spatial summation** (5.3) · **summation** (5.3) · **synapse** (5.1) · **synaptic pruning** (5.5) · **synaptic strength** (5.1) · **temporal summation** (5.3) · **threshold** (5.3)

REVIEW QUESTIONS

1. What was at stake in the dispute between the neuron doctrine and the view of the nervous system as a continuous web?
2. State Hebb's postulate in your own words, and explain why long-term potentiation was taken as evidence for it.
3. Explain why the action potential is called all-or-none, and how a neuron nevertheless signals the strength of its input.
4. What did the comparison between Thorndike's line-drawers and Trowbridge and Cason's informed groups show?
5. Describe the evidence that infants lose the ability to distinguish speech sounds their language does not use.

FURTHER READING

Kandel, E. R. (2006). *In search of memory: The emergence of a new science of mind.* W. W. Norton.

LeDoux, J. (2002). *Synaptic self: How our brains become who we are.* Viking.

CHAPTER 6

Bias

THE PLACARD BY THE STAIRS · 6F

The ways a mind leans, and the arguments that only look sound.

The syllabus is on the desk.

Learning Objectives

1. Define a cognitive bias and distinguish it from random error.
2. Explain the framing effect, with medical and public-policy examples.
3. Describe anchoring and explain why it persists when the anchor is known to be arbitrary.
4. Describe confirmation bias and biased assimilation of evidence.
5. Explain hindsight bias, and identify common informal fallacies in argument.

A **cognitive bias** is a systematic departure from the judgment that the evidence warrants. The word that matters is *systematic*. Random error scatters in every direction and tends to cancel out over many judgments; a bias leans one way, so that more judgments make it more visible rather than less. Herbert Simon (1955) argued that human reasoning is **boundedly rational**: it works with limited time, information and attention, and so relies on shortcuts. In a landmark paper, Amos Tversky and Daniel Kahneman (1974) described several such shortcuts, which they called **heuristics**, and showed that each produces characteristic biases. This chapter describes five of the best documented.

6.1 Framing

ON THE BOARD · 6F BIAS · **FRAMING**

Ninety in a hundred survive. Ten in a hundred die.

It is the same number. People choose differently between them.

The way a thing is put is doing the choosing.

The board's example comes from medicine. McNeil, Pauker, Sox and Tversky (1982) asked patients, physicians and students to choose between surgery and radiation therapy for lung cancer. Some were told the outcomes of surgery in terms of survival (of 100 people having surgery, 90 survive the treatment period); others were told the same outcomes in terms of mortality (10 die). Surgery was considerably more popular when its outcomes were described as survival, and the effect was as strong among physicians as among patients. The numbers were identical. The description chose.

This is the **framing effect**: different but logically equivalent descriptions of the same options lead to different choices. Tversky and Kahneman (1981) gave the best-known example. Participants imagined that an unusual disease was expected to kill 600 people and chose between two programmes. When the programmes were described in terms of lives saved (200 people will be saved for certain, or a one-third chance that all 600 will be saved), most chose the certain option. When the same programmes were described in terms of deaths (400 people will die for certain, or a two-thirds chance that all 600 will die), most chose the gamble.

Kahneman and Tversky's **prospect theory** (1979) explains the pattern. People evaluate outcomes as gains or losses relative to a reference point, and a frame sets that point. They tend to avoid risk when choosing among gains and to seek risk when choosing among losses, and losses weigh more heavily than equivalent gains, a property known as **loss aversion**.

6.2 Anchors

ON THE BOARD · 6F BIAS · **ANCHORS**

The first number heard pulls every guess after it toward itself.

Even when everyone knows it was picked at random.

RESEARCH SPOTLIGHT · THE WHEEL OF FORTUNE

Tversky and Kahneman (1974) spun a wheel of fortune, numbered from 0 to 100, in front of participants. In the version Kahneman (2011) later described, the wheel was rigged to stop at either 10 or 65. Participants were asked whether the percentage of African countries in the United Nations was higher or lower than the number on the wheel, and then to estimate the actual percentage. Participants who had seen 10 gave a median estimate of 25 per cent; those who had seen 65 gave a median of 45. A number everyone had seen chosen by chance had pulled every estimate toward itself.

Anchoring is the tendency for an initial value to bias subsequent numerical estimates toward it. Tversky and Kahneman attributed it to **insufficient adjustment**: people start from the anchor and adjust away from it, but stop too soon. Later research added a second mechanism, **selective accessibility**: considering whether the answer could be as high (or as low) as the anchor brings to mind information consistent with that possibility (Strack & Mussweiler, 1997).

Expertise does not protect against anchoring. Northcraft and Neale (1987) found that estate agents' valuations of a house were pulled toward the listing price they had been given, though most denied that the price had influenced them. Englich, Mussweiler and Strack (2006) found that experienced judges' sentencing decisions in a hypothetical case were influenced by a sentencing demand they had determined themselves by throwing dice. The board is right on both counts: the first number pulls, and it pulls even when everyone knows it was picked at random.

6.3 Confirming

ON THE BOARD · 6F BIAS · **CONFIRMING**

People look for what agrees with them, and find it.

What disagrees gets checked twice as hard.

Confirmation bias is the tendency to seek, interpret and remember information in ways that support what one already believes (Nickerson, 1998). Peter Wason studied it with two famous tasks.

TRY IT · THE 2-4-6 TASK

The sequence *2, 4, 6* follows a rule. Propose other sequences of three numbers, and for each you will be told whether it fits the rule. When you are confident, state the rule. (Have a friend play the role of Wason, with the rule written down in advance: *any three numbers in increasing order*.) In Wason's (1960) study, most participants proposed sequences such as *8, 10, 12* and *20, 22, 24*, each of which fitted their hypothesis of "numbers increasing by two", received a yes each time, and announced the wrong rule with confidence. Few tried a sequence that their hypothesis predicted would fail, such as *1, 2, 3*, which would have revealed the truth.

In Wason's **selection task** (introduced in 1966; Wason, 1968), participants see four cards and must decide which to turn over to test a rule such as "if a card has a vowel on one side, it has an even number on the other." Most choose the cards that could confirm the rule and neglect the one that could refute it. Only a small minority choose correctly.

Confirmation bias also shapes how evidence is weighed. Lord, Ross and Lepper (1979) gave supporters and opponents of capital punishment the same two studies, one apparently supporting and one apparently undermining its deterrent effect. Each side rated the study that agreed with it as more convincing, found flaws in the other, and ended more confident in its original view. This **biased assimilation** is the board's second sentence: what disagrees is checked harder. Ditto and Lopez (1992) found that people required more evidence before accepting an unwelcome conclusion than a welcome one, a pattern they called **motivated scepticism**.

6.4 Hindsight

ON THE BOARD · 6F BIAS · **HINDSIGHT**

Once you know how it ended, it looks as if it had to end that way.

It did not.

Hindsight bias is the tendency, once an outcome is known, to see it as having been more predictable than it was. Baruch Fischhoff (1975) gave participants an account of a little-known historical conflict, the war between the British and the Gurkhas of Nepal in 1814, and told different groups different outcomes. Whichever outcome a group was told, its members judged that outcome to have been the most likely, and believed they would have predicted it. Fischhoff called this **creeping determinism**: knowledge of the outcome creeps into the reconstruction of what could have been foreseen.

Hindsight bias affects memory as well as judgment. Fischhoff and Beyth (1975) asked people to estimate the probabilities of possible outcomes of President Nixon's visits to China and the Soviet Union before the visits took place. Afterwards, asked to recall their estimates, they remembered having given higher probabilities to the events that had occurred.

The bias matters wherever decisions are judged after the fact. A good decision can turn out badly, and a bad one well; judging decisions by their outcomes, **outcome bias**, punishes the first and rewards the second (Baron & Hershey, 1988). A useful corrective is to **consider the opposite**: before concluding that an outcome was inevitable, list the ways it could have gone differently (Lord, Lepper & Preston, 1984).

6.5 Fallacies

ON THE BOARD · 6F BIAS · **FALLACIES**

A fallacy is an argument that looks as if it works.

Attacking the speaker. Two choices where there are more.

Everybody does it, so it is fine.

A **fallacy** is an argument that seems valid or persuasive but is not. Aristotle catalogued fallacies in *On Sophistical Refutations*, and the subject has been studied ever since (Hamblin, 1970). **Formal fallacies** are errors in an argument's logical structure. The board lists three **informal fallacies**, which are errors of content or relevance.

- **Ad hominem** ("to the person"): attacking the person who makes an argument rather than the argument. Whether a claim is true does not depend on the character of the person who states it.
- **False dilemma** (or false dichotomy): presenting two options as the only ones when others exist. "Either we ban it or we accept chaos."
- **Appeal to popularity** (*argumentum ad populum*, the bandwagon fallacy) and its close relation, the **appeal to common practice**: concluding that something is true or acceptable because many people believe or do it.

Recognising fallacies is easier in other people's arguments than in one's own. Pronin, Lin and Ross (2002) found that people readily rated others as more susceptible to a range of biases than they were themselves, a pattern they called the **bias blind spot**. Learning the names of biases does not, by itself, remove them.

A CLOSER LOOK · ARE BIASES ALWAYS ERRORS?

Some researchers argue that heuristics are often well suited to the environments in which they are used, and that laboratory tasks can make them look worse than they are (Gigerenzer, 1996). Klayman and Ha (1987) showed that testing cases one expects to succeed, the strategy that fails in the 2-4-6 task, is often an effective strategy when rules are narrow. The debate is about how to interpret the biases, not about whether the effects occur. The effects described in this chapter have been replicated many times.

CHAPTER SUMMARY

- A bias is a systematic, not random, departure from what the evidence warrants; heuristics are useful shortcuts that produce characteristic biases (intro).
- Logically equivalent descriptions of the same outcomes lead to different choices; prospect theory explains the pattern by reference points and loss aversion (6.1).
- Initial numbers anchor later estimates, even when known to be random and even for experts (6.2).
- People test beliefs by seeking confirmation and scrutinise disagreeing evidence more harshly (6.3).
- Known outcomes seem to have been predictable, distorting both judgment and memory (6.4).
- Informal fallacies such as ad hominem, false dilemma and appeal to popularity are easier to see in others than in oneself (6.5).

KEY TERMS

ad hominem (6.5) · **anchoring** (6.2) · **appeal to common practice** (6.5) · **appeal to popularity** (6.5) · **bias blind spot** (6.5) · **biased assimilation** (6.3) · **bounded rationality** (intro) · **cognitive bias** (intro) · **confirmation bias** (6.3) · **consider the opposite** (6.4) · **creeping determinism** (6.4) · **fallacy** (6.5) · **false dilemma** (6.5) · **formal fallacy** (6.5) · **framing effect** (6.1) · **heuristic** (intro) · **hindsight bias** (6.4) · **informal fallacy** (6.5) · **insufficient adjustment** (6.2) · **loss aversion** (6.1) · **motivated scepticism** (6.3) · **outcome bias** (6.4) · **prospect theory** (6.1) · **selection task** (6.3) · **selective accessibility** (6.2)

REVIEW QUESTIONS

1. Why is a bias more visible across many judgments, while random error is less?
2. Restate the two versions of the disease problem and explain the reversal of preference using prospect theory.
3. Why are the dice and wheel-of-fortune studies especially telling evidence of anchoring?
4. Explain why most participants in the 2-4-6 task failed to discover the rule.
5. Distinguish hindsight bias from outcome bias.
6. Identify the fallacy: "You cannot trust her figures on traffic; she does not even drive."

FURTHER READING

Kahneman, D. (2011). *Thinking, fast and slow*. Farrar, Straus and Giroux.

Nickerson, R. S. (1998). Confirmation bias: A ubiquitous phenomenon in many guises. *Review of General Psychology, 2*(2), 175-220.

CHAPTER 7

Error

THE PLACARD BY THE STAIRS · 7F

Being wrong, and why that is where the change happens.

The last floor. The syllabus is on the desk.

Learning Objectives

1. Explain the idea that perception and learning are driven by prediction, and define prediction error.
2. State the principle of error-driven learning and describe blocking as evidence for it.
3. Explain why the size of each correction matters, and describe the consequences of overcorrection.
4. Describe evidence that errors followed by feedback improve learning.
5. Distinguish confidence from accuracy, and describe overconfidence and calibration.

The last floor of the building teaches that being wrong is where the change happens. The idea has a long history and a precise modern form. The mind continually anticipates what will happen next; when what happens differs from what was expected, the difference, the **prediction error**, is the signal that drives learning. When nothing unexpected happens, there is nothing to learn.

7.1 Surprise

ON THE BOARD · 7F ERROR · **SURPRISE**

A mind is always guessing what comes next.

Surprise is the size of the miss.

In the nineteenth century, Hermann von Helmholtz (1867) argued that perception involves **unconscious inference**: the brain does not simply receive sensations but interprets them, drawing conclusions about their causes from past experience. Contemporary **predictive processing** accounts develop the idea. On these accounts the brain continually generates expectations about incoming signals and passes forward chiefly the part of the signal it did not expect, the error (Clark, 2013).

The brain's response to the unexpected can be measured directly. In the **oddball paradigm**, a listener hears a regular sequence of identical tones interrupted occasionally by a different one. Even when the listener is reading a book and ignoring the sounds, the deviant tone evokes a distinctive electrical response, the **mismatch negativity**, first described by Näätänen, Gaillard and Mäntysalo (1978). The brain registers the violation of its expectation whether or not it is attending.

The board's definition is exact: surprise is the size of the miss. The bigger the gap between what was expected and what occurred, the greater the surprise, and, as the next section shows, the greater the learning.

7.2 Correction

ON THE BOARD · 7F ERROR · **CORRECTION**

Learning is mostly correcting.

The bigger the miss, the bigger the correction.

In 1972 Robert Rescorla and Allan Wagner proposed a rule for how animals learn to associate a cue with an outcome, which became one of the most influential ideas in the psy-

chology of learning. The **Rescorla-Wagner rule** states that the change in the strength of an association on each trial is proportional to the prediction error: the difference between the outcome that occurred and the outcome that all present cues predicted.

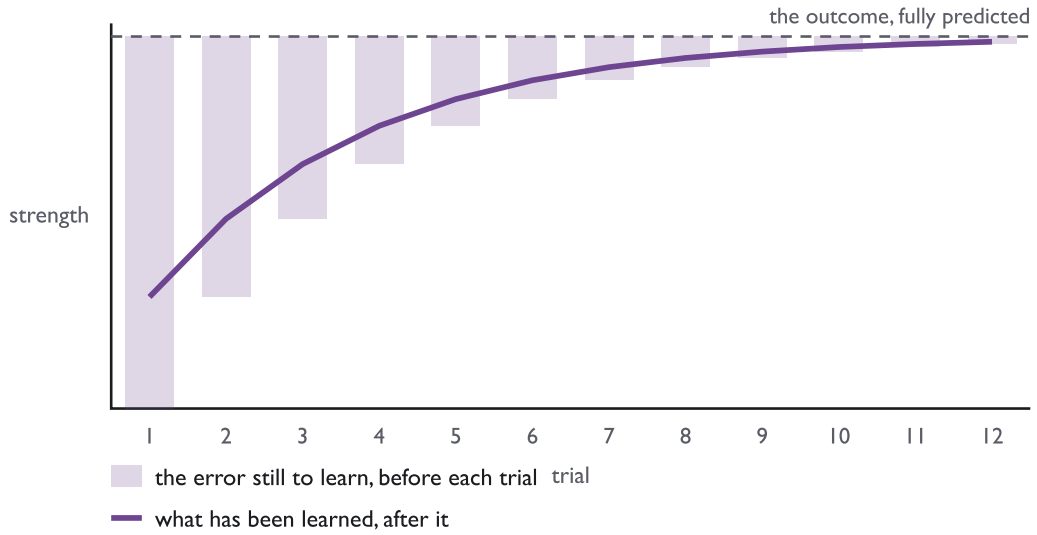


Figure 7.1 Error-driven learning. On early trials the outcome is unexpected and the error is large, so the association grows quickly. As the outcome becomes predicted, the error shrinks and learning slows.

The rule explains why learning curves are steep at first and then level off: early on, the outcome is surprising and each trial produces a large correction; later, the outcome is expected and each trial changes little. It also explains a striking phenomenon discovered by Leon Kamin (1969).

RESEARCH SPOTLIGHT · BLOCKING

Kamin first trained rats to expect a mild shock after a noise. He then presented the noise together with a light, followed by the same shock. Finally he tested the light alone. The rats showed almost no learning about the light, although it had been paired with the shock many times. The noise already predicted the shock fully, so the shock was not surprising, and without surprise there was no learning about the new cue. This **blocking** effect is exactly what an error-driven rule predicts, and what a rule based on mere pairing does not.

Neurophysiology later found a signal with the same properties. Schultz, Dayan and Montague (1997) reported that dopamine neurons in the monkey brain fire in response to an unexpected reward. Once a cue reliably predicted the reward, the neurons fired to the cue and no longer to the reward itself; and when an expected reward failed to arrive, their activity dipped below its usual level at the moment the reward should have come. The neurons were signalling not reward but **reward prediction error**.

7.3 Steps

ON THE BOARD · 7F ERROR · **STEPS**

Correct too much at once and you overshoot.

Small corrections, many times, get closer.

The Rescorla-Wagner rule includes a factor that sets how much of each error is corrected on a single trial, which in animal learning reflects the salience of the cue and the outcome. If that factor is small, learning is slow but steady. If it is large, each trial produces a large change, and when outcomes vary from trial to trial, large changes chase each fluctuation, overshooting in one direction and then the other. Correcting by a fraction of each error, many times, settles closer to the truth than correcting the whole error at once.

The same principle appears wherever people adjust to feedback. In **prism adaptation**, studied by Helmholtz himself, a person wearing prisms that shift the visual field to one side at first points to the side of targets, then corrects gradually over many attempts. When the prisms are removed, the person errs in the opposite direction, an **aftereffect** that shows the correction had been absorbed into the movement itself, and that it, too, must be undone gradually.

Teaching uses small steps deliberately. In **shaping**, described by B. F. Skinner (1953), a complex behaviour is built by reinforcing **successive approximations**, each a small step closer to the goal, rather than waiting for the complete behaviour to appear.

A CLOSER LOOK · CORRECTING FOR LUCK

Overcorrection is especially likely when an outcome is partly due to chance. Kahneman (2011) describes flight instructors who believed that praise for an excellent landing was followed by a worse one, and criticism for a poor landing by a better one. In fact both followed from **regression to the mean**: an unusually good or bad performance is usually followed by a more ordinary one, whatever is said in between. A learner who makes a large correction after every unusually bad result is correcting for luck, and will overshoot.

7.4 Mistakes

ON THE BOARD · 7F ERROR · **MISTAKES**

A mistake you notice teaches more than an answer you were given.

It seems natural that making mistakes during learning would be harmful, since the mistake might be remembered instead of the right answer. A body of research suggests otherwise, provided the mistake is followed by correction. In the **generation effect**, information that learners produce themselves is remembered better than information they simply read (Slamecka & Graf, 1978). Kornell, Hays and Bjork (2009) went further: participants who attempted to guess the answer to a question before being shown it remembered the answer better than those who studied the answer for the same total time, even when their guesses were wrong. Attempting an answer before studying, **pretesting**, improves learning of the material that follows (Richland, Kornell & Kao, 2009).

The brain registers its own mistakes rapidly. When people make an error in a speeded task, a distinctive electrical signal, the **error-related negativity**, appears within about a tenth of a second of the incorrect response, often before the person is aware of having erred (Falkenstein et al., 1991; Gehring et al., 1993).

Noticing is the condition. A mistake that is never noticed, because no feedback arrives, teaches nothing; Chapter 5 showed that practice without knowledge of results produces habit, not skill. A mistake that is noticed carries exactly the information a given answer does not: where the learner's own expectation was wrong, and by how much (Metcalfe, 2017).

7.5 Certainty

ON THE BOARD · 7F ERROR · **CERTAINTY**

Being sure is not evidence.

Nothing surprises the surest, so they learn the least.

Confidence is a feeling about a judgment, not a measurement of it. A person is well **calibrated** if their confidence matches their accuracy: of all the answers they are 70 per cent sure of, about 70 per cent are right. Research on calibration, reviewed by Lichtenstein, Fischhoff and Phillips (1982), found that people are usually **overconfident**, especially on difficult questions. In one classic procedure, participants give a range they are 98 per cent sure contains the true value of some quantity; their ranges miss the true value far more often than two per cent of the time (Alpert & Raiffa, 1982).

Overconfidence is not confined to novices. Philip Tetlock (2005) collected thousands of predictions about political and economic events from experts over many years. Their accuracy was modest, and the experts who were most certain, those who explained everything through one big idea, were the least accurate; those who drew on many ideas and qualified their judgments did better. Kruger and Dunning (1999) reported that people with the least skill in a domain tended to overestimate their performance most, because the skill needed to perform well is also the skill needed to recognise good performance; the size and interpretation of that effect are still debated.

The board's final sentence follows from the chapter. If learning is driven by surprise, and surprise is the gap between expectation and outcome, a person who is certain will seek no test of their belief and will register no error, and so will not learn. Interestingly, when highly confident people *are* shown to be wrong, the large surprise makes the correction especially memorable, a **hypercorrection effect** (Butterfield & Metcalfe, 2001). Certainty does not prevent learning from a correction; it prevents looking for one.

CHAPTER SUMMARY

- The brain continually predicts its input, and responds to violations of prediction even without attention (7.1).
- Learning is proportional to prediction error, which explains the shape of learning curves, blocking, and the firing of dopamine neurons (7.2).
- Correcting a fraction of each error, repeatedly, converges better than correcting all of it at once, especially when outcomes include chance (7.3).
- Errors followed by feedback improve learning; the brain registers errors within a tenth of a second (7.4).
- Confidence and accuracy are distinct; people are commonly overconfident, and certainty discourages the testing that learning requires (7.5).

KEY TERMS

aftereffect (7.3) · **blocking** (7.2) · **calibration** (7.5) · **error-related negativity** (7.4) · **generation effect** (7.4) · **hypercorrection effect** (7.5) · **mismatch negativity** (7.1) · **oddball paradigm** (7.1) · **overconfidence** (7.5) · **prediction error** (intro, 7.1) · **predictive processing** (7.1) · **pretesting** (7.4) · **prism adaptation** (7.3) · **regression to the mean** (7.3) · **Rescorla-Wagner rule** (7.2) · **reward prediction error** (7.2) · **shaping** (7.3) · **successive approximation** (7.3) · **unconscious inference** (7.1)

REVIEW QUESTIONS

1. What is the mismatch negativity, and why is it evidence that the brain predicts its input?
2. State the Rescorla-Wagner rule in words, and use it to explain blocking.
3. Describe the activity of dopamine neurons before and after a cue comes to predict a reward.

4. Why might large corrections slow learning when outcomes are partly due to chance?
5. Under what conditions do errors help learning?
6. Distinguish confidence from calibration, and explain why certainty can prevent learning.

FURTHER READING

Metcalfe, J. (2017). Learning from errors. *Annual Review of Psychology, 68*, 465-489.

Schulz, K. (2010). *Being wrong: Adventures in the margin of error.* Ecco.

Conclusion: What Happens to a Piece of Information

The Seven Floors Together

Each chapter of this book has treated one subject, as each floor of the school does. Read in order, they describe a single journey.

A piece of information is first **named**. Language gives it a handle, assembled from pieces, set in an order, and fixed to one meaning by its context (Chapter 1). It is then **noticed**, or not: attention, limited and directed in advance, selects a small part of what is present and lets the rest pass unseen (Chapter 2). What is noticed may be **kept**, not as a copy but as materials for rebuilding, which fade, compete with one another, and change with each recall (Chapter 3). What is kept is **sorted** into categories that have middles but no clean edges, at levels of generality, by features that some theory made relevant (Chapter 4). Experience **changes** the brain itself, join by join, as connections used together strengthen and those unused are removed (Chapter 5). Judgment is **bent** by the way options are framed, by anchors, by the wish to be right, and by knowledge of how things turned out (Chapter 6). And finally the whole system is **corrected**, because the gap between what was expected and what occurred is the signal that drives learning (Chapter 7).

The chapters also speak to one another. Names shape what is remembered (1.1, 3.1). Attention determines what can be learned (2.5, 7.4). Recall rebuilds memory, and confidence in a rebuilt memory can outrun its accuracy (3.4, 7.5). Categories decide which features are noticed (4.3, 2.3). Feedback is the condition of skill (5.4) for the same reason that prediction error is the engine of learning (7.2). And biases persist partly because they rarely produce the noticed error that would correct them (6.5, 7.5).

Studying for the Examination

The school publishes its examination's terms in advance. Each floor's syllabus states that the examination covers the board and that the pass mark is 80 out of 100. The research in this book suggests how to prepare.

1. **Space your study** (3.5). Visit each chapter's boards on several days rather than all at once.
2. **Practise retrieval** (3.5). Close the book and recall each board in your own words before checking it.
3. **Attempt before you look** (7.4). Answer each chapter's review questions before rereading the chapter.
4. **Mind the alike** (3.3). Topics that resemble one another, such as proactive and retroactive interference, are the most easily confused; study them side by side and name the difference.
5. **Check your calibration** (7.5). For each answer you give in practice, note how sure you are, and see whether your confidence matches your accuracy.
6. **Welcome the wrong answers** (7.2, 7.4). A noticed mistake shows you exactly where your expectation was wrong. That is the most useful thing a practice test can give you.

A Last Word

The board on the top floor ends with the warning that nothing surprises the surest, so they learn the least. It is a fitting end to an introduction. Every finding in this book was once a surprise to someone who expected otherwise and took the trouble to check. The habit of checking, of treating one's own certainty as a hypothesis, is the most durable thing a course on the mind can teach.

Glossary

action potential The brief, all-or-none electrical signal a neuron sends along its axon when its membrane potential crosses threshold. (5.3)

ad hominem A fallacy that attacks the person making an argument instead of the argument. (6.5)

agent The thematic role of the one who performs an action. (1.3)

anchoring The pull of an initial value on subsequent numerical estimates. (6.2)

arbitrariness The absence of any natural resemblance between a word's form and its meaning. (1.1)

attention The processes by which some information is selected for further processing at the expense of other information. (Ch. 2)

attenuation Treisman's proposal that unattended input is weakened rather than blocked. (2.1)

basic level The level of a taxonomy (e.g., chair) at which categories are most informative for the least effort. (4.4)

bias blind spot The tendency to see biases in others more readily than in oneself. (6.5)

biased assimilation Evaluating evidence as stronger when it supports one's prior view. (6.3)

blocking The failure to learn about a new cue when the outcome is already predicted by another cue. (7.2)

bottleneck A processing stage that can serve only one task at a time. (2.4)

calibration The match between a person's confidence and their accuracy. (7.5)

change blindness The failure to notice a change to a scene, especially across an interruption. (2.5)

cognitive bias A systematic departure from the judgment that the evidence warrants. (Ch. 6)

confirmation bias The tendency to seek, interpret and remember information in ways that support existing beliefs. (6.3)

conjunction search Search for a target defined by a combination of features; time grows with set size. (2.3)

cross-modal priming A method in which a spoken word's effect on recognition of a visually presented word is measured. (1.4)

deliberate practice Practice designed to improve specific weaknesses, with immediate feedback. (5.4)

desirable difficulties Conditions, such as spacing, that make learning feel harder but make it last longer. (3.5)

encoding Taking information in and transforming it into a form that can be stored. (Ch. 3)

endogenous cue A cue that directs attention voluntarily, such as an arrow. (2.2)

error-related negativity An electrical brain signal that follows an erroneous response within about 100 milliseconds. (7.4)

exemplar theory The view that a category is represented by stored individual examples. (4.1)

exogenous cue A cue that captures attention automatically, such as a sudden flash. (2.2)

fallacy An argument that seems valid or persuasive but is not. (6.5)

false dilemma A fallacy that presents two options as the only ones when others exist. (6.5)

family resemblance A pattern of overlapping similarities among category members, with no feature shared by all. (Ch. 4)

feature search Search for a target differing from distractors in a single feature; it pops out. (2.3)

forgetting curve The relation between time since learning and amount retained: steep at first, then levelling. (3.2)

framing effect Different choices produced by logically equivalent descriptions of the same options. (6.1)

generation effect Better memory for information produced by the learner than for information read. (7.4)

Hebb's postulate The proposal that a connection strengthens when one cell repeatedly takes part in firing another. (5.2)

heuristic A mental shortcut that usually works but produces characteristic biases. (Ch. 6)

hindsight bias Seeing an outcome, once known, as more predictable than it was. (6.4)

hypercorrection effect The finding that high-confidence errors, once corrected, are especially well corrected. (7.5)

inattentional blindness Failure to notice a visible but unexpected object while attending elsewhere. (2.5)

interference Impairment of memory by other learning; proactive (old on new) or retroactive (new on old). (3.3)

law of effect Thorndike's principle that responses followed by satisfying outcomes become more likely. (5.4)

levels of processing The proposal that deeper, meaning-based processing produces better memory. (3.1)

long-term potentiation A lasting increase in synaptic strength following intense activity. (5.2)

loss aversion The tendency for losses to weigh more heavily than equivalent gains. (6.1)

misinformation effect The incorporation of information encountered after an event into memory of the event. (3.4)

mismatch negativity The brain's electrical response to a deviant sound in a regular sequence. (7.1)

morpheme The smallest unit of language that carries meaning. (1.2)

neuron A nerve cell; the discrete unit of the nervous system. (5.1)

orthographic depth The consistency with which a writing system maps sounds to letters. (1.5)

overconfidence Confidence that exceeds accuracy. (7.5)

permastore Bahrick's term for knowledge that survives for decades after initial decline. (3.2)

polysemy Related senses carried by a single word. (1.4)

pop-out The immediate detection of a target that differs in a single feature. (2.3)

prediction error The difference between what was expected and what occurred. (Ch. 7)

productivity The capacity of language to produce and understand new expressions from known parts. (1.2)

prospect theory Kahneman and Tversky's account of choice in terms of gains and losses from a reference point. (6.1)

prototype A summary representation of a category's central tendency. (4.1)

reconsolidation The restabilising of a memory after retrieval, during which it can be modified. (3.4)

reconstructive memory The view that remembering rebuilds an event from stored materials. (3.1)

regression to the mean The tendency of an extreme outcome to be followed by a less extreme one. (7.3)

Rescorla-Wagner rule The principle that associative learning on each trial is proportional to prediction error. (7.2)

residual category A category for what fits no other, such as *other* or *miscellaneous*. (4.5)

retrieval Bringing stored information back into use. (Ch. 3)

schema An organised body of knowledge used to interpret and reconstruct experience. (3.1)

sensitive period A period in development during which experience has an especially strong effect. (5.5)

shaping Building a behaviour by reinforcing successive approximations to it. (7.3)

spacing effect Better long-term retention from study distributed over time than from massed study. (3.5)

summation The adding up of inputs to a neuron, across time (temporal) or across synapses (spatial). (5.3)

switch cost The extra time required when changing from one task to another. (2.4)

synapse The junction at which one neuron passes a signal to another. (5.1)

synaptic pruning The developmental elimination of unused synapses. (5.5)

synaptic strength How large a change one firing of a sending cell produces in a receiving cell. (5.1)

syntax The rules by which words combine into phrases and sentences. (1.3)

testing effect Better retention from practising recall than from re-studying. (3.5)

threshold The membrane potential at which a neuron fires an action potential. (5.3)

typicality How good an example of its category an item is judged to be. (4.1)

unconscious inference Helmholtz's term for the interpretation of sensations by drawing on past experience. (7.1)

References

Alpert, M., & Raiffa, H. (1982). A progress report on the training of probability assessors. In D. Kahneman, P. Slovic, & A. Tversky (Eds.), *Judgment under uncertainty: Heuristics and biases* (pp. 294-305). Cambridge University Press.

Azevedo, F. A. C., et al. (2009). Equal numbers of neuronal and nonneuronal cells make the human brain an isometrically scaled-up primate brain. *Journal of Comparative Neurology, 513*(5), 532-541.

Bahrick, H. P. (1984). Semantic memory content in permastore: Fifty years of memory for Spanish learned in school. *Journal of Experimental Psychology: General, 113*(1), 1-29.

Baron, J., & Hershey, J. C. (1988). Outcome bias in decision evaluation. *Journal of Personality and Social Psychology, 54*(4), 569-579.

Bartlett, F. C. (1932). *Remembering: A study in experimental and social psychology.* Cambridge University Press.

Berko, J. (1958). The child's learning of English morphology. *Word, 14*(2-3), 150-177.

Bever, T. G. (1970). The cognitive basis for linguistic structures. In J. R. Hayes (Ed.), *Cognition and the development of language* (pp. 279-362). Wiley.

Bjork, R. A. (1994). Memory and metamemory considerations in the training of human beings. In J. Metcalfe & A. Shimamura (Eds.), *Metacognition: Knowing about knowing* (pp. 185-205). MIT Press.

Bliss, T. V. P., & Lømo, T. (1973). Long-lasting potentiation of synaptic transmission in the dentate area of the anaesthetized rabbit following stimulation of the perforant path. *Journal of Physiology, 232*(2), 331-356.

Bowker, G. C., & Star, S. L. (1999). *Sorting things out: Classification and its consequences.* MIT Press.

Broadbent, D. E. (1958). *Perception and communication.* Pergamon Press.

Butterfield, B., & Metcalfe, J. (2001). Errors committed with high confidence are hypercorrected. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 27*(6), 1491-1494.

Carmichael, L., Hogan, H. P., & Walter, A. A. (1932). An experimental study of the effect of language on the reproduction of visually perceived form. *Journal of Experimental Psychology, 15*(1), 73-86.

Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). Distributed practice in verbal recall tasks: A review and quantitative synthesis. *Psychological Bulletin, 132*(3), 354-380.

Cherry, E. C. (1953). Some experiments on the recognition of speech, with one and with two ears. *Journal of the Acoustical Society of America, 25*(5), 975-979.

Chomsky, N. (1957). *Syntactic structures.* Mouton.

Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences, 36*(3), 181-204.

Collins, A. M., & Quillian, M. R. (1969). Retrieval time from semantic memory. *Journal of Verbal Learning and Verbal Behavior, 8*(2), 240-247.

Craik, F. I. M., & Lockhart, R. S. (1972). Levels of processing: A framework for memory research. *Journal of Verbal Learning and Verbal Behavior, 11*(6), 671-684.

Craik, F. I. M., & Tulving, E. (1975). Depth of processing and the retention of words in episodic memory. *Journal of Experimental Psychology: General, 104*(3), 268-294.

Ditto, P. H., & Lopez, D. F. (1992). Motivated skepticism: Use of differential decision criteria for preferred and nonpreferred conclusions. *Journal of Personality and Social Psychology, 63*(4), 568-584.

Ebbinghaus, H. (1885). *Über das Gedächtnis.* Duncker & Humblot. (Trans. 1913 as *Memory: A contribution to experimental psychology.*)

Englich, B., Mussweiler, T., & Strack, F. (2006). Playing dice with criminal sentences: The influence of irrelevant anchors on experts' judicial decision making. *Personality and Social Psychology Bulletin, 32*(2), 188-200.

Ericsson, K. A., Krampe, R. T., & Tesch-Römer, C. (1993). The role of deliberate practice in the acquisition of expert performance. *Psychological Review, 100*(3), 363-406.

Falkenstein, M., Hohnsbein, J., Hoormann, J., & Blanke, L. (1991). Effects of crossmodal divided attention on late ERP components. II. Error processing in choice reaction tasks. *Electroencephalography and Clinical Neurophysiology, 78*(6), 447-455.

Fischhoff, B. (1975). Hindsight is not equal to foresight: The effect of outcome knowledge on judgment under uncertainty. *Journal of Experimental Psychology: Human Perception and Performance, 1*(3), 288-299.

Fischhoff, B., & Beyth, R. (1975). "I knew it would happen": Remembered probabilities of once-future things. *Organizational Behavior and Human Performance, 13*(1), 1-16.

Fitts, P. M., & Posner, M. I. (1967). *Human performance.* Brooks/Cole.

Foucault, M. (1966). *Les mots et les choses.* Gallimard. (Trans. 1970 as *The order of things.*)

Frege, G. (1892). Über Sinn und Bedeutung. *Zeitschrift für Philosophie und philosophische Kritik, 100*, 25-50.

Gehring, W. J., Goss, B., Coles, M. G. H., Meyer, D. E., & Donchin, E. (1993). A neural system for error detection and compensation. *Psychological Science, 4*(6), 385-390.

Gigerenzer, G. (1996). On narrow norms and vague heuristics: A reply to Kahneman and Tversky. *Psychological Review, 103*(3), 592-596.

Hamblin, C. L. (1970). *Fallacies.* Methuen.

Hebb, D. O. (1949). *The organization of behavior: A neuropsychological theory.* Wiley.

Helmholtz, H. von (1867). *Handbuch der physiologischen Optik.* Voss.

Hockett, C. F. (1960). The origin of speech. *Scientific American, 203*(3), 88-96.

Hodgkin, A. L., & Huxley, A. F. (1952). A quantitative description of membrane current and its application to conduction and excitation in nerve. *Journal of Physiology, 117*(4), 500-544.

Huttenlocher, P. R. (1979). Synaptic density in human frontal cortex: Developmental changes and effects of aging. *Brain Research, 163*(2), 195-205.

Huttenlocher, P. R., & Dabholkar, A. S. (1997). Regional differences in synaptogenesis in human cerebral cortex. *Journal of Comparative Neurology, 387*(2), 167-178.

James, W. (1890). *The principles of psychology.* Henry Holt.

Jenkins, J. G., & Dallenbach, K. M. (1924). Obliviscence during sleep and waking. *American Journal of Psychology, 35*(4), 605-612.

Jersild, A. T. (1927). Mental set and shift. *Archives of Psychology, 89.*

Kahneman, D. (1973). *Attention and effort.* Prentice-Hall.

Kahneman, D. (2011). *Thinking, fast and slow.* Farrar, Straus and Giroux.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica, 47*(2), 263-291.

Kamin, L. J. (1969). Predictability, surprise, attention, and conditioning. In B. A. Campbell & R. M. Church (Eds.), *Punishment and aversive behavior* (pp. 279-296). Appleton-Century-Crofts.

Katz, L., & Frost, R. (1992). The reading process is different for different orthographies: The orthographic depth hypothesis. In R. Frost & L. Katz (Eds.), *Orthography, phonology, morphology, and meaning* (pp. 67-84). North-Holland.

Klayman, J., & Ha, Y.-W. (1987). Confirmation, disconfirmation, and information in hypothesis testing. *Psychological Review, 94*(2), 211-228.

Kluger, A. N., & DeNisi, A. (1996). The effects of feedback interventions on performance: A historical review, a meta-analysis, and a preliminary feedback intervention theory. *Psychological Bulletin, 119*(2), 254-284.

Kornell, N. (2009). Optimising learning using flashcards: Spacing is more effective than cramming. *Applied Cognitive Psychology, 23*(9), 1297-1317.

Kornell, N., Hays, M. J., & Bjork, R. A. (2009). Unsuccessful retrieval attempts enhance subsequent learning. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 35*(4), 989-998.

Kruger, J., & Dunning, D. (1999). Unskilled and unaware of it: How difficulties in recognizing one's own incompetence lead to inflated self-assessments. *Journal of Personality and Social Psychology, 77*(6), 1121-1134.

Labov, W. (1973). The boundaries of words and their meanings. In C.-J. N. Bailey & R. W. Shuy (Eds.), *New ways of analyzing variation in English* (pp. 340-373). Georgetown University Press.

Levin, D. T., Momen, N., Drivdahl, S. B., & Simons, D. J. (2000). Change blindness blindness: The metacognitive error of overestimating change-detection ability. *Visual Cognition, 7*(1-3), 397-412.

Lichtenstein, S., Fischhoff, B., & Phillips, L. D. (1982). Calibration of probabilities: The state of the art to 1980. In D. Kahneman, P. Slovic, & A. Tversky (Eds.), *Judgment under uncertainty: Heuristics and biases* (pp. 306-334). Cambridge University Press.

Loftus, E. F., & Palmer, J. C. (1974). Reconstruction of automobile destruction: An example of the interaction between language and memory. *Journal of Verbal Learning and Verbal Behavior, 13*(5), 585-589.

Loftus, E. F., & Pickrell, J. E. (1995). The formation of false memories. *Psychiatric Annals, 25*(12), 720-725.

Lord, C. G., Lepper, M. R., & Preston, E. (1984). Considering the opposite: A corrective strategy for social judgment. *Journal of Personality and Social Psychology, 47*(6), 1231-1243.

Lord, C. G., Ross, L., & Lepper, M. R. (1979). Biased assimilation and attitude polarization: The effects of prior theories on subsequently considered evidence. *Journal of Personality and Social Psychology, 37*(11), 2098-2109.

Mack, A., & Rock, I. (1998). *Inattentional blindness.* MIT Press.

Maguire, E. A., Gadian, D. G., Johnsrude, I. S., Good, C. D., Ashburner, J., Frackowiak, R. S. J., & Frith, C. D. (2000). Navigation-related structural change in the hippocampi of taxi drivers. *Proceedings of the National Academy of Sciences, 97*(8), 4398-4403.

McCloskey, M. E., & Glucksberg, S. (1978). Natural categories: Well defined or fuzzy sets? *Memory & Cognition, 6*(4), 462-472.

McGeoch, J. A. (1932). Forgetting and the law of disuse. *Psychological Review, 39*(4), 352-370.

McNeil, B. J., Pauker, S. G., Sox, H. C., & Tversky, A. (1982). On the elicitation of preferences for alternative therapies. *New England Journal of Medicine, 306*(21), 1259-1262.

Medin, D. L., & Schaffer, M. M. (1978). Context theory of classification learning. *Psychological Review, 85*(3), 207-238.

Metcalfe, J. (2017). Learning from errors. *Annual Review of Psychology, 68*, 465-489.

Moray, N. (1959). Attention in dichotic listening: Affective cues and the influence of instructions. *Quarterly Journal of Experimental Psychology, 11*(1), 56-60.

Murphy, G. L., & Medin, D. L. (1985). The role of theories in conceptual coherence. *Psychological Review, 92*(3), 289-316.

Murre, J. M. J., & Dros, J. (2015). Replication and analysis of Ebbinghaus' forgetting curve. *PLOS ONE, 10*(7), e0120644.

Näätänen, R., Gaillard, A. W. K., & Mäntysalo, S. (1978). Early selective-attention effect on evoked potential reinterpreted. *Acta Psychologica, 42*(4), 313-329.

Nader, K., Schafe, G. E., & LeDoux, J. E. (2000). Fear memories require protein synthesis in the amygdala for reconsolidation after retrieval. *Nature, 406*, 722-726.

Nagy, W. E., & Anderson, R. C. (1984). How many words are there in printed school English? *Reading Research Quarterly, 19*(3), 304-330.

Nickerson, R. S. (1998). Confirmation bias: A ubiquitous phenomenon in many guises. *Review of General Psychology, 2*(2), 175-220.

Northcraft, G. B., & Neale, M. A. (1987). Experts, amateurs, and real estate: An anchoring-and-adjustment perspective on property pricing decisions. *Organizational Behavior and Human Decision Processes, 39*(1), 84-97.

Pashler, H. (1994). Dual-task interference in simple tasks: Data and theory. *Psychological Bulletin, 116*(2), 220-244.

Posner, M. I. (1980). Orienting of attention. *Quarterly Journal of Experimental Psychology, 32*(1), 3-25.

Pronin, E., Lin, D. Y., & Ross, L. (2002). The bias blind spot: Perceptions of bias in self versus others. *Personality and Social Psychology Bulletin, 28*(3), 369-381.

Rescorla, R. A., & Wagner, A. R. (1972). A theory of Pavlovian conditioning: Variations in the effectiveness of reinforcement and nonreinforcement. In A. H. Black & W. F. Prokasy (Eds.), *Classical conditioning II* (pp. 64-99). Appleton-Century-Crofts.

Richland, L. E., Kornell, N., & Kao, L. S. (2009). The pretesting effect: Do unsuccessful retrieval attempts enhance learning? *Journal of Experimental Psychology: Applied, 15*(3), 243-257.

Rips, L. J., Shoben, E. J., & Smith, E. E. (1973). Semantic distance and the verification of semantic relations. *Journal of Verbal Learning and Verbal Behavior, 12*(1), 1-20.

Roediger, H. L., & Karpicke, J. D. (2006). Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science, 17*(3), 249-255.

Rogers, R. D., & Monsell, S. (1995). Costs of a predictable switch between simple cognitive tasks. *Journal of Experimental Psychology: General, 124*(2), 207-231.

Rosch, E. (1975). Cognitive representations of semantic categories. *Journal of Experimental Psychology: General, 104*(3), 192-233.

Rosch, E., & Mervis, C. B. (1975). Family resemblances: Studies in the internal structure of categories. *Cognitive Psychology, 7*(4), 573-605.

Rosch, E., Mervis, C. B., Gray, W. D., Johnson, D. M., & Boyes-Braem, P. (1976). Basic objects in natural categories. *Cognitive Psychology, 8*(3), 382-439.

Sachs, J. S. (1967). Recognition memory for syntactic and semantic aspects of connected discourse. *Perception & Psychophysics, 2*(9), 437-442.

Saussure, F. de (1916). *Cours de linguistique générale.* Payot.

Schultz, W., Dayan, P., & Montague, P. R. (1997). A neural substrate of prediction and reward. *Science, 275*(5306), 1593-1599.

Seymour, P. H. K., Aro, M., & Erskine, J. M. (2003). Foundation literacy acquisition in European orthographies. *British Journal of Psychology, 94*(2), 143-174.

Shatz, C. J. (1992). The developing brain. *Scientific American, 267*(3), 60-67.

Simon, H. A. (1955). A behavioral model of rational choice. *Quarterly Journal of Economics, 69*(1), 99-118.

Simons, D. J., & Chabris, C. F. (1999). Gorillas in our midst: Sustained inattentional blindness for dynamic events. *Perception, 28*(9), 1059-1074.

Simons, D. J., & Levin, D. T. (1998). Failure to detect changes to people during a real-world interaction. *Psychonomic Bulletin & Review, 5*(4), 644-649.

Skinner, B. F. (1953). *Science and human behavior.* Macmillan.

Slamecka, N. J., & Graf, P. (1978). The generation effect: Delineation of a phenomenon. *Journal of Experimental Psychology: Human Learning and Memory, 4*(6), 592-604.

Strack, F., & Mussweiler, T. (1997). Explaining the enigmatic anchoring effect: Mechanisms of selective accessibility. *Journal of Personality and Social Psychology, 73*(3), 437-446.

Strayer, D. L., & Johnston, W. A. (2001). Driven to distraction: Dual-task studies of simulated driving and conversing on a cellular telephone. *Psychological Science, 12*(6), 462-466.

Swinney, D. A. (1979). Lexical access during sentence comprehension: (Re)consideration of context effects. *Journal of Verbal Learning and Verbal Behavior, 18*(6), 645-659.

Talarico, J. M., & Rubin, D. C. (2003). Confidence, not consistency, characterizes flashbulb memories. *Psychological Science, 14*(5), 455-461.

Tanaka, J. W., & Taylor, M. (1991). Object categories and expertise: Is the basic level in the eye of the beholder? *Cognitive Psychology, 23*(3), 457-482.

Tetlock, P. E. (2005). *Expert political judgment: How good is it? How can we know?* Princeton University Press.

Thorndike, E. L. (1898). Animal intelligence: An experimental study of the associative processes in animals. *Psychological Review Monograph Supplements, 2*(4).

Thorndike, E. L. (1931). *Human learning.* Century.

Treisman, A. M. (1960). Contextual cues in selective listening. *Quarterly Journal of Experimental Psychology, 12*(4), 242-248.

Treisman, A. M., & Gelade, G. (1980). A feature-integration theory of attention. *Cognitive Psychology, 12*(1), 97-136.

Trowbridge, M. H., & Cason, H. (1932). An experimental study of Thorndike's theory of learning. *Journal of General Psychology, 7*(2), 245-260.

Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science, 185*(4157), 1124-1131.

Tversky, A., & Kahneman, D. (1981). The framing of decisions and the psychology of choice. *Science, 211*(4481), 453-458.

Underwood, B. J. (1957). Interference and forgetting. *Psychological Review, 64*(1), 49-60.

Wason, P. C. (1960). On the failure to eliminate hypotheses in a conceptual task. *Quarterly Journal of Experimental Psychology, 12*(3), 129-140.

Wason, P. C. (1968). Reasoning about a rule. *Quarterly Journal of Experimental Psychology, 20*(3), 273-281.

Werker, J. F., & Tees, R. C. (1984). Cross-language speech perception: Evidence for perceptual reorganization during the first year of life. *Infant Behavior and Development, 7*(1), 49-63.

Wickens, D. D., Born, D. G., & Allen, C. K. (1963). Proactive inhibition and item similarity in short-term memory. *Journal of Verbal Learning and Verbal Behavior, 2*(5-6), 440-445.

Wittgenstein, L. (1953). *Philosophical investigations.* Blackwell.

Wixted, J. T., & Ebbesen, E. B. (1991). On the form of forgetting. *Psychological Science, 2*(6), 409-415.

Wolfe, J. M. (1994). Guided Search 2.0: A revised model of visual search. *Psychonomic Bulletin & Review, 1*(2), 202-238.

Zhang, W., & Linden, D. J. (2003). The other side of the engram: Experience-driven changes in neuronal intrinsic excitability. *Nature Reviews Neuroscience, 4*(11), 885-900.

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